

Economics Coauthorships in the Aftermath of MeToo

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Abstract

We study changes in coauthorships in economics, after the MeToo movement, using NBER and CEPR working papers between January 2004 and December 2020. We identify three main shifts in collaboration patterns. First, compared to pre-MeToo levels, collaborations across genders and an author's seniority group increased: we estimate a 12.3% increase of women coauthors per 100 men-authored papers. Second, coauthorship shares of senior with junior economists declined by 3.0%, indicating a shift towards sorting of collaborations by seniority. Third, shares of new coauthorships declined by 5.4%, driven by drops in senior economists' shares of new junior and new junior women by 18.4% and 48.0% respectively. The results are robust to different specifications.

JEL Classification: J16, J24, J44

Keywords: MeToo, gender, coauthorship patterns, networks.

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1 Introduction

Scientific collaboration is a fundamental aspect of research as it facilitates the exchange of ideas, knowledge, data, and method sharing. Importantly, it builds intergenerational links, with senior researchers playing a key role in mentoring young scholars by offering guidance, support, and chances to broaden their networks.¹ Any change to collaboration incentives can influence researchers' productivity, job prospects, and career advancement, with varying effects based on career stage and the size and strength of coauthorship networks. The MeToo movement gained international momentum in October 2017, and it has increased awareness around issues of gender discrimination and harassment in the workplace since then. In 2018, economists faced their own MeToo moment with increasingly abundant evidence of harassment and discriminatory culture in the profession.² Institutions and associations have responded with measures to address these issues, and the debate has extended across the academic community, social media, and institutions. This event could have significantly influenced the nature and dynamics of academic collaborations. The MeToo movement's impact on the economics profession may have fostered a more respectful research environment, increased scrutiny of existing practices, and promoted greater diversity and inclusivity within the community. Conversely, it could have induced a chilling effect on collaborations, potentially causing researchers to become more hesitant in forming partnerships outside their established networks due to heightened concerns about trust and reputational risk.

This paper examines changes in coauthorship dynamics within economics following the MeToo movement. We analyse these changes from the perspective of individual authors, investigating shifts in coauthor composition by seniority and gender. Our approach distinguishes between pre-existing and newly established coauthorship relationships, allowing us to study both the effects of the movement on the expansion of economists' professional networks and the evolution of established collaborations. We identify three key changes post-MeToo: First, collaborations across genders within an author's seniority group increased. Second, collaborations between senior and junior economists declined. Third, the formation of new coauthorship relationships decreased, particularly between senior and junior researchers.

Our analysis considers 26,791 working papers published in either the Centre for Economic Policy Research (CEPR) discussion paper series, or the National Bureau of Economic Research (NBER) working paper series, two prominent outlets for economics research. These were written by 17,998 distinct authors from January 2004 to December 2020.³ For each paper, we observe the title, publication date, JEL codes, and author names. We complement these data with rich information about each author using details from CEPR and NBER, as well as from publicly available CVs, LinkedIn profiles, and personal or organisational webpages. These sources allow us to assign gender and seniority at the time of publication, proxy professional experience using PhD graduation date, and construct characteristics of authors' coauthorship networks. We define seniority categories based on their positions at the time of publication as early-career (e.g. Postdoctoral Fellow), junior (e.g., Assistant Professors), midcareer (e.g., Associate Professors), and senior (e.g., Full Professors, Chaired

¹The literature highlights a growing trend of collaboration in the economics profession (Ginther and Kahn (2004), Hamermesh (2013)), with teams having more authors and gender diversity producing more impactful work (Anderson and Richards-Shubik (2022), Maddi and Gingras (2021)). Ductor, Goyal and Prummer (2023) notes differences in collaboration networks across genders.

²The MeToo movement in economics was discussed at the *American Economic Association* annual meeting in January 2019, and the AEA moved to establish processes to handle complaints and has produced several reports on the professional climate in economics (see for example <https://19thnews.org/2022/10/me-too-movement-economics-twitter-misconduct/>, <https://www.nytimes.com/2020/01/07/business/economy/economics-race-gender.html>, <https://www.nytimes.com/2019/01/10/business/economics-sexual-harassment-metoo.html>, and <https://www.aeaweb.org/news/member-announcements-sept-26-2019>). The results of these reports and the measures that have been taken were recently discussed in a panel session at the ASSA 2023 Annual Meeting in New Orleans: <https://www.aeaweb.org/webcasts/2023/harassment-lessons-learned>.

³The dataset we use in this paper is a subset of the data set collected for *Project CAPER* used in Amano-Patiño, Faraglia, Giannitsarou and Hasna (2020). See <https://sites.google.com/view/projectcaper/home> for details.

Professors), acknowledging that our sample may overrepresent established scholars due to the selective nature of CEPR and NBER memberships.

We proxy the onset of the MeToo movement in the economics profession as the end of the second quarter of 2018. To quantify shifts in coauthor composition, we use a logistic regression framework. This approach allows us to estimate the relationship between the post-MeToo period and the probability of collaborating with coauthors of different seniorities and genders; and in addition, the likelihood of forming new collaborations within these categories.

Our findings reveal three significant shifts in collaboration patterns compared to pre-MeToo averages. First, we estimate a rise in the proportion of women coauthors for men, both overall and within junior and senior subgroups. Conversely, we find a decrease in the proportion of women coauthors for women, both overall and within corresponding seniority levels. Using a back-of-the-envelope calculation, these increases in mixed-gender collaborations, translate to an estimated 12.3% increase in women coauthors per 100 men-authored papers. Second, we estimate decreases in the proportion of senior coauthors (especially senior women) for juniors, and symmetrically, in the proportion of junior coauthors (particularly junior women) for seniors. The decreases in collaborations between senior and junior economists we quantify, suggest a 3.0% decrease in the share of senior authors collaborating with junior coauthors. These first two findings indicate that post-MeToo, authors have increasingly sorted their collaborations by seniority rather than by gender. Third, we find evidence of a general chilling effect on the expansion of economists' professional networks. We estimate decreases in the share of new coauthors across all seniorities and genders, the share of new senior coauthors for juniors, and the share of new junior coauthors for seniors. Our estimates translate into 5.4% fewer new coauthorships per 100 papers. This trend is primarily driven by a substantial decrease in new coauthorships between senior and junior authors: for seniors, the share of new junior coauthors has dropped by 18.4%, with a particularly sharp 48% decrease in their share of new junior women coauthors.

Next, we use several strategies to rule out alternative explanations that we could be mistakenly attributing to the MeToo movement. First, we conduct a placebo test that replaces the MeToo period with an earlier 'placebo' period of the same duration (July 1, 2014, to December 31, 2016). Reassuringly, we find small or no effects on any of the outcomes we study during this placebo period. Second, we leverage the heterogeneity in the expected time it takes to write different types of working papers, which creates variation in the timing in which a paper's collaboration team is formed and therefore, how affected it is by MeToo even if the publication date is fixed. While this analysis has limitations — notably, that it is challenging to precisely determine the average time from inception to working paper draft, and that papers often undergo changes in their authorship team over time — our results are stronger when we exclude slower papers, mitigating the concern that our estimates could confound MeToo effects from pre-existing patterns or trends.

To validate our choice of the MeToo starting date, we estimate our model multiple times, varying the beginning of the post-MeToo period from October 1, 2017, to December 1, 2019, in three month increments. This approach yields estimates for different starting dates of the MeToo indicator, that support our choice of the end of the second quarter of 2018 as the starting point, even though the exact MeToo starting date is difficult to pinpoint precisely. We address potential gender-related confounding factors by interacting author-level characteristics with the sex indicator. The results from this specification remain consistent with our main findings. Additionally, we corroborate that our findings are robust to different model specifications and definitions of new coauthorships.

Lastly, we explore heterogeneity across institutions, namely NBER vs. CEPR, and field groups, namely applied micro, macro, and theory. Interestingly, our results across institutions remain consistent but show

differences that may reflect distinct approaches to mentorship and collaboration in (mostly) European versus US academic contexts. Our analysis across field groups uncovers significant heterogeneity. However, our sample size limits our ability to conduct a more granular analysis at the field group level, constraining our insights to these broader field groups.

All combined, our findings suggest a slowdown in collaboration network expansion across genders and seniorities, potentially influenced by the MeToo movement.

This study contributes to the existing research on academic researchers' unequal experiences and career paths. Early studies by [Ginther and Kahn \(2004\)](#), [Bayer and Rouse \(2016\)](#) and [Card and DellaVigna \(2013\)](#) examined collaboration and publishing in academia, highlighting gender diversity issues in economics. Recent research reveals productivity disparities between male and female economists, identifying obstacles women face from graduate program entry to promotion and publication ([Adams and Lowry \(2022\)](#); [Card, DellaVigna, Funk and Iriberry \(2020\)](#); [Chari and Goldsmith-Pinkham \(2017\)](#); [Dupas, Modestino, Niederle, Wolfers et al. \(2021\)](#); [Ghosh and Liu \(2020\)](#); [Heckman and Moktan \(2020\)](#); [Hengel \(2022\)](#); [Huang, Gates, Sinatra and Barabási \(2020\)](#); [Kosnik \(2022\)](#); [Lusher, Yang and Carrell \(2023\)](#); [Bansak, Meade and Starr-McCluer \(2022\)](#); [Paredes, Paserman and Pino \(2020\)](#); [Sarsons \(2017\)](#); [Sarsons, Gërkhani, Reuben and Schram \(2021\)](#)).⁴ Our research is related to [Gertsberg \(2024\)](#), who explores the impact of the MeToo movement on junior academic women's collaborations in top US economics departments, revealing a decline in partnerships with senior colleagues post-2018. We complement these findings with a comprehensive dataset covering 26,791 papers and 17,998 authors from various career stages and institutions globally. Additionally, we contribute to the literature on the Covid-19 pandemic's effects on research productivity, complementing studies by [Amano-Patiño et al. \(2020\)](#) and [Deryugina et al. \(2021\)](#), and others, which indicate reduced research time, productivity, and submission rates for women academics during the pandemic's early stages. Our findings also support those of [Kruger et al. \(2022\)](#), suggesting that the pandemic led to an increased reliance on past collaborations, with greater productivity gains for more central authors in the collaboration network.

The paper is organised as follows. [Section 2](#) describes the data and the variables used in the empirical analysis. [Section 3](#) outlines our empirical strategy and our results. [Section 4](#) presents robustness checks and [Section 5](#) further results by publication outlet and fields. [Section 6](#) concludes.

2 Data

We assess how research collaboration and coauthorship patterns in economics have changed pre- and post-MeToo using working papers published in the NBER and CEPR. We view coauthorship from the perspective of *an author in a paper* and examine how the shares of coauthors in a given paper, by gender and seniority differ pre- and post-MeToo. For this purpose, we leverage a subsample of the dataset collected for *Project CAPER* and used in [Amano-Patiño et al. \(2020\)](#), which includes all working papers published in the NBER Working Paper series and the CEPR Discussion Paper series between January 2004 and December 2020. For each of these, we have the title, publication date, JEL codes (where available), and author names. We complemented the data to incorporate author characteristics, such as years of professional experience, number of past coauthors, number of previous papers, average number of coauthors per paper, and measures of the size of their collaboration network, as represented by their involvement in papers published in the two series.

Our raw data consists of the set of distinct papers published in either the NBER or the CEPR between

⁴An extensive summary of the literature can be found in the AEA *Status of Women in the Economics Profession*, <https://www.aeaweb.org/about-aea/committees/cswep/survey/related-literature>.

January 2004 and December 2020. Some papers appear at different stages of development in both these outlets, so for this reason, we use the fuzzy matching method developed by [Amano-Patiño, Contractor and Aramburu \(2024\)](#) on titles and sets of authors, to remove ‘duplicate’ papers, where a paper has been submitted at different times to different working paper series, while we retain information on each outlet a paper appears in. We keep the first submission for each paper, since this most accurately reflects when coauthorships develop.

To control for a possible time trend of increased research outputs in these two outlets, as well as author characteristics that may be changing over time, we split the sample of papers published between 1 January 2005 and 31 December 2020, into eight two-year periods, e.g. period $t = 1$ includes papers from 1 January 2005 to 31 December 2006, period $t = 2$ includes papers from 1 January 2007 to 31 December 2008, etc. The data and papers from the pre-period $t = 0$, from the 1 January 2004 to 31 December 2004 are used to initialize lagged variables of interest for authors.

The total number of author-paper observations from 1 January 2004 to 31 December 2020, is 67,142. The number of distinct authors is 17,998 and the number of distinct papers is 26,791.⁵

2.1 NBER and CEPR Working Paper Outlets

The CEPR and NBER are leading nonprofit organisations that advance economic research through networks of affiliated scholars. These institutions play a crucial role in rapidly disseminating their members’ newest research, typically taking only about one week from submission to publication of working papers. Our data comprises these members and their coauthors. To provide context for our sample, we briefly describe the institutional characteristics of these organisations.

Each network has over 1,700 members. The selection process for both organisations is highly competitive, taking place annually for NBER, and biennially for CEPR.

NBER affiliates must be associated with a US-based academic institution. Selection begins each February with research group-specific nominations. Program directors, assisted by leading scholars, review candidates and appoint new members. NBER maintains a two-tier system: tenured scholars become Research Associates, while untenured scholars are appointed as Faculty Research Fellows. Approximately 80% of NBER’s affiliates hold Research Associate status. Each year, a small number of inactive researchers are removed.

CEPR appointments to specific research groups are decided by a committee comprising of the CEPR President, Vice-President for Appointments, and relevant Programme Directors. While members can be based outside Europe, they must demonstrate interest in European economics and travel to Europe occasionally. CEPR also maintains a two-tier system: fellows are established scholars, while affiliates are promising researchers who completed their PhD within the past seven years. Fellow appointments last four years and are reviewed based on CEPR activity participation, with exceptions for career breaks. Affiliates may be promoted to Fellow status after their four-year term, based on academic contributions and engagement. Inactive researchers are not reappointed. CEPR fellows can nominate only one potential new affiliate and one new fellow per cycle.

Given the selective nature of NBER and CEPR memberships, it is likely that our sample over-represents established scholars, skewing observed coauthorship patterns. This could particularly affect our analysis of junior-senior collaborations, as junior economists face higher entry barriers to these networks. However, both institutions recognise these barriers and reserve memberships for promising early-career scholars, which can alleviate some of the senior over-representation in our sample. Additionally, the inclusion in our sample of coauthors who are not network members helps mitigate this bias.⁶

⁵We have excluded papers with seven or more coauthors, which constitute less than 0.5% of our observations.

⁶All the previous information was provided directly by Professor James Poterba (President of NBER) and Professor Beatrice

2.2 Author-paper Variables

We define a variable of coauthorship for author i in paper p with author j as

$$ca_{ip}(j) = \begin{cases} 1 & \text{if author } j \text{ is an author in paper } p, \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where we set $ca_{ip}(i) = 1$ for each paper, even if i is the only author of the paper, to include single-authored papers in the sample, whilst being consistent in the treatment of coauthorships. This means that all author-paper level characteristics include a ‘self-link’.⁷ For each coauthorship $ca_{ip}(j)$ we have the date of first submission of the paper, and use this to work out coauthorships of author i in each period. We then use author-paper-coauthor variables to calculate the author-paper level characteristics, which serve as the basis for our dependent variables. For an author-paper pair (i, p) , the first set of dependent variables of interest are the *shares of coauthors* in paper p by gender, seniority and gender-seniority, generally denoted by y_{ip} . For each of these, we also define its corresponding *relevant yearly share* $r(y_{ip}, t)$ as the average of y_{ip} across all papers in our sample published in year t , where t is the year paper p was published. For example, when y_{ip} corresponds to the share of senior coauthors in author i ’s paper p , then $r(y_{ip}, t)$ is the average share of senior coauthors across all papers published in year t .⁸

The setup also allows us to identify the proportion of *new* coauthorships in each paper and define the second set of dependent variables of interest as the shares of new coauthors for author i in paper p again by gender, seniority and gender-seniority. A coauthorship of author i with author j in paper p is defined as *new*, if there has been no other paper dated before paper p in which the two authors collaborated in our data. If after the first time we observe a coauthorship, additional papers coauthored by the same two authors appear within a three-month interval, we count these additional coauthorships as new too, since it is not possible to determine which paper was really the first to be initiated by the two authors.

2.3 Author characteristics

For each author in our sample, we identify a set of characteristics and gather information about their career paths using details from the CEPR and NBER, as well as publicly available data from CVs, LinkedIn profiles, and personal or organizational webpages. For example, we manually assign gender to each author based on their first names and pictures, and we extract information on PhD completion year, job title and institution of affiliation at the time of paper publication from these sources. We use the PhD completion year to define the *experience* of an author i in observation (i, p) as the number of years since PhD completion at the time of publication of paper p . In our period-arranged data, experience is defined as the time elapsed between the middle of the period and the year of PhD graduation. We also use PhD year and PhD institution information to ensure that we uniquely identify each author, despite variations in name spelling.

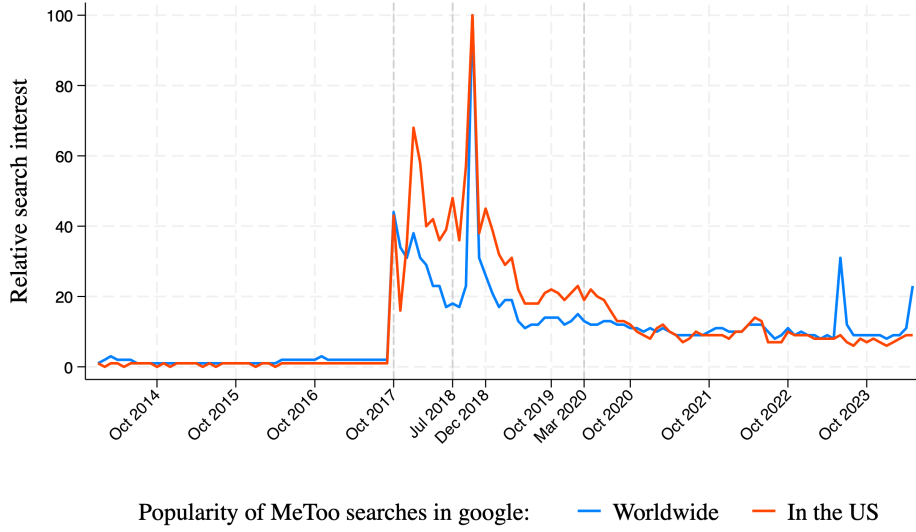
Additional author-level controls in our regressions include the total number of papers published in the CEPR or NBER series by the author, up until the time period that the paper belongs to, and the number of papers written by the author in the same period as that of the paper of the given observation. These last two

Weder di Mauro (President of CEPR) in response to inquiries from the authors.

⁷In [Appendix A.2](#) we provide a simple constructed example using a toy dataset containing four papers and three authors to illustrate the construction of coauthorship and related variables.

⁸This relevant share variable is included as a control in our regressions to disentangle changes in coauthorship patterns attributable to MeToo from broader temporal trends in author composition.

Figure 1: MeToo searches in the US and worldwide



Notes: The figure shows the relative search interest from Google Trends, scaled from 0 to 100, with 100 being the peak popularity for the term. Google Trends does not provide absolute search volume numbers.

measures serve as proxies for output and productivity of authors.

We include two author control variables related to the network of coauthorships of authors. The first is the lagged coauthorship *degree* of an author, which measures the number of distinct coauthors the author had collaborated with until the last day in period $t - 1$ for an author-paper observation in period t . The second is lagged *clustering* of an author, which captures the extent to which the author clusters, or collaborates, together with a group of other authors, again until the last day in period $t - 1$ for an author-paper observation in period t .⁹

Finally, we define author's *seniority* using the author i 's title/position at the date that paper p from observation (i, p) appears in our dataset. We use five seniority categories: senior, mid-career, junior, early-career and unclassified. For academics, *senior* is assigned to Professors, Full Professors or Chaired positions, *midcareer* is assigned to Associate Professors, Senior Lecturers or Readers, *junior* is assigned to non-tenured Assistant Professors and Lecturers, and *early-career* is assigned to all remaining non-established researchers, i.e. Post-docs, PhD candidates, Pre-docs, Masters' and Undergraduate students. For non-academic researchers, we use standard metrics of seniority.¹⁰ The remaining category, *unclassified*, is used for all authors who could not be easily put into one of the other classifications.

2.4 Paper Characteristics

The end of 2017 marks the beginning of the post-MeToo period across the world, as evidenced by the sharp increase in Google search interest shown in Figure 1. This figure illustrates the global impact and rapid spread of the MeToo movement, with search interest peaking in late 2017 and early 2018. To account for the possible lag between initiating new research projects and the time it takes to submit them in the two working paper series we study, we create a dummy variable $metoo_p$, which equals 1 if paper p is dated after the end of the second quarter of 2018 onwards, and 0 otherwise. This timing allows for the movement's impact to manifest

⁹To construct these, we follow the corresponding definitions of Ductor et al. (2023), and provide details in the Section A.1.

¹⁰We have contacted the most prevalent non academic institutions that appear in our dataset, such as Federal Reserve Banks, ECB, other central banks in Europe, the IMF, the World Bank, the WTO, etc. and asked them to assign every position to the above-outlined seniority scale. For other institutions we use standard positions.

in the economics literature. We defer a detailed discussion on the time threshold for MeToo to [Section 4.3.1](#).

We also define the variable $covid_p$ as the indicator of paper p being dated after the end of the first quarter of 2020. This variable helps us distinguish between the effects of the MeToo movement and the subsequent disruptions caused by the COVID-19 pandemic on research collaborations and publishing patterns.

In addition to the above, we use the following control variables related to a paper in observation (i, p) : the total number of authors in a paper, including author i , and a set of dummy variables for each of the 20 JEL code letters, to control for the field of the paper.¹¹

2.5 Samples and statistics

We create three versions of the sample. The first sample includes *all authors* and papers as described in the previous section. To avoid artificially inflating the number of new coauthorships, we remove the first appearance of each author in the sample, as all coauthorships in their initial paper are new, by construction. This adjustment also removes any new coauthorships defined by the three-month cutoff in [Section 2.2](#), as well as observations for which the author appears only once with a single paper in the whole sample. The remaining sample consists of 46,449 author-paper observations, with 8,117 distinct authors and 24,240 distinct papers. The second and third samples are subsets of the *all authors* sample, containing only author-paper observations for *junior* and *senior authors*, respectively, based on our definition of seniority at the time of submission to NBER or CEPR.¹²

[Table 1](#) provides summary statistics of the datasets we use for our analysis. All our subsamples are composed of mostly men authors (73–83%). Our data slightly underrepresents women compared to overall journal publications in economics, primarily because NBER and CEPR members tend to be more senior researchers. However, the share of women in our sample follows a trend consistent with the findings of [Ductor and Prummer \(2024\)](#), as illustrated in [Figure B.1](#). This trend suggests that despite the underrepresentation, our data captures the broader shift towards increased women’s participation in economics research.

Men have on average larger coauthorship networks and women have on average more clustered collaboration networks. More senior authors have more coauthors and less clustered networks and, as expected, are more experienced than the average author from the full sample. These authors’ characteristics are in line with what the literature finds, e.g. [Ductor et al. \(2023\)](#).

As [Figure B.2](#) shows, the number of papers published annually in either NBER or CEPR from 2004 to 2020 shows an upward trend, with a notable spike in 2020. While year-to-year changes typically ranged between -7% and 11% , the increase from 2019 to 2020 was exceptionally high at 25.03% , even excluding Covid-related papers. This surge underscores the unusual productivity during the pandemic’s onset and the importance of accounting for it. Concurrently, and shown in the same [Figure B.2](#), the share of single-author papers has steadily declined from 20% in 2004 to less than 10% in 2020, highlighting the growing importance of collaborations in economics research over our sample period. These trends reflect both the increasing output in these prestigious working paper series and a shift towards more collaborative research practices in the field.

[Figure 2](#) illustrates that all-men papers have decreased over time, from over 70% in 2004 to an average of 56% by 2020. Notably, despite an overall increase in the proportion of women in our sample over time (as shown in [Figure B.1](#)), the share of women-only authored papers has remained relatively stable, oscillating

¹¹Some papers in the CEPR discussion series since 2015 have not been assigned a JEL code, and these are dropped from our analysis.

¹²We do not consider the *midcareer* subsample due to the limited number of midcareer women, which prevents a meaningful analysis. Similarly, the *early career* subsample is very small to do any meaningful analysis.

Table 1: Summary Statistics

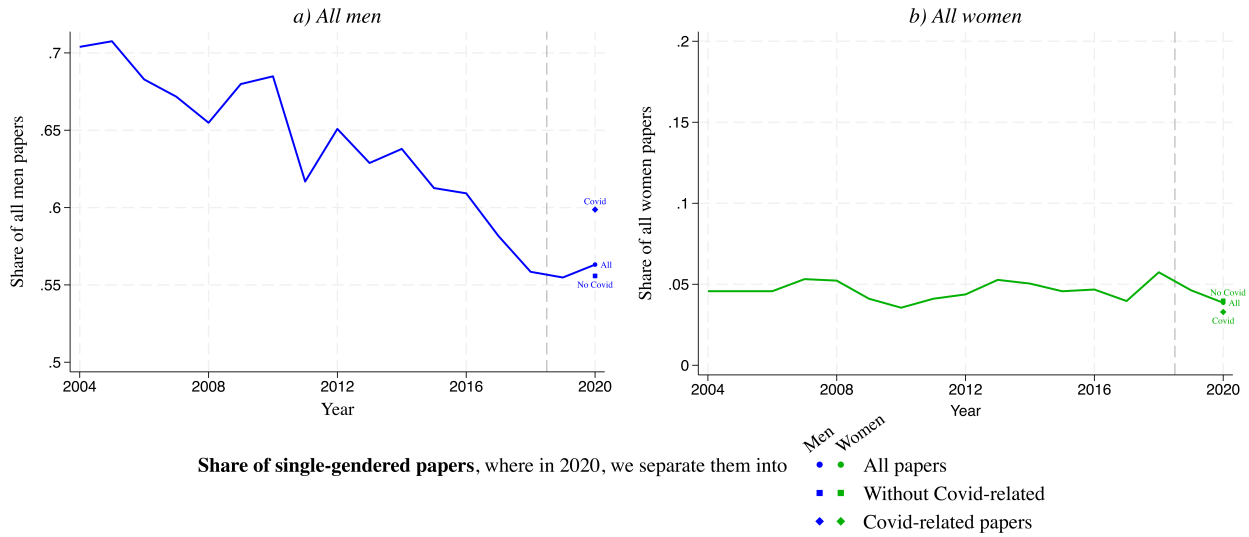
	All	Men	Women
<i>All authors</i>			
Number of papers	24240		
Number of authors	8117	6308	1809
Authors' average characteristics			
Lagged degree	4.38	4.53	3.83
Lagged clustering	0.85	0.84	0.88
Years of experience	12.10	12.80	9.66
Number of coauthors per paper	3.02	3.01	3.03
Number of papers per period	2.01	2.06	1.82
Cumulative num. of papers	5.07	5.27	4.18
Authors' seniorities at publication			
Early career economists	495	346	149
Junior economists	2319	1677	642
Micareer economists	1790	1351	439
Senior economists	3255	2747	508
Unclassified	258	187	71
<i>Junior authors</i>			
Number of papers	7425		
Number of authors	3234	2375	859
Authors' average characteristics			
Lagged degree	3.10	3.17	2.91
Lagged clustering	0.92	0.91	0.93
Years of experience	5.17	5.08	5.41
Number of coauthors per paper	2.94	2.93	2.94
Number of papers per period	1.99	2.02	1.89
Cumulative num. of papers	3.62	3.70	3.41
Authors' seniorities at publication			
Early career economists	0	0	0
Junior economists	3234	2375	859
Micareer economists	0	0	0
Senior economists	0	0	0
Unclassified	0	0	0
<i>Senior authors</i>			
Number of papers	19126		
Number of authors	3690	3090	600
Authors' average characteristics			
Lagged degree	6.66	6.69	6.49
Lagged clustering	0.74	0.73	0.75
Years of experience	19.66	20.10	17.44
Number of coauthors per paper	2.98	2.96	3.04
Number of papers per period	2.28	2.32	2.09
Cumulative num. of papers	7.87	7.99	7.27
Authors' seniorities at publication			
Early career economists	0	0	0
Junior economists	0	0	0
Micareer economists	0	0	0
Senior economists	3690	3090	600
Unclassified	0	0	0

Notes: Number of authors by seniority in *All authors* sample is counted using the mode seniority of each distinct author. Average characteristics are calculated as the average across the mean characteristic of distinct authors.

around 5% throughout the period. This stability suggests that the increase in mixed-gendered papers is not solely attributable to a greater availability of women coauthors.

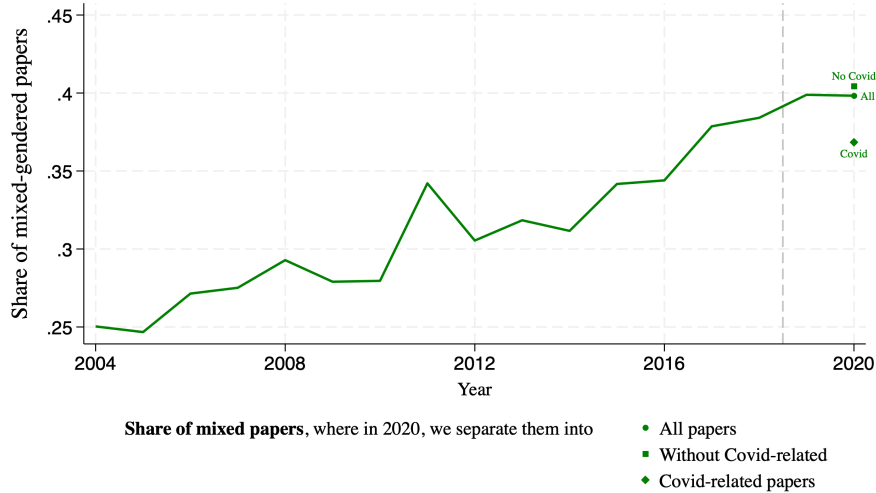
As men represent the majority of authors in our sample, the share of mixed-gender papers has increased over time, mirroring the declining trend in all-men papers, as shown in Figure 3. This trend likely reflects changing collaboration patterns among male authors, who are increasingly working with female colleagues.

Figure 2: Share of Single-Gendered Papers by Year



Notes: The graphs show the shares of single-gendered papers published year by year in either NBER or CEPR from 2004–2020. Note that the y-axis are at a different scale with all-men papers being over 10 times that of all-women papers.

Figure 3: Share of Mixed-Gendered Papers by Year



Notes: The figure shows the share of mixed-gendered papers published year by year in either NBER or CEPR from 2004–2020.

3 Analysis

To investigate changes in coauthorship patterns pre- and post-MeToo, we examine the composition of coauthors in paper p from the perspective of author i . Our dependent variables are functions of shares y_{ip} of coauthors

of specific types, such as the share of women coauthors or women of particular seniority levels, where the unit of observation is an author-paper pair (i, p) . We consider specifications using both overall shares of coauthors and shares of *new* coauthors, allowing us to study changes in both overall and new coauthorship patterns post-MeToo.

Our preferred specifications follow a logistic regression model where our dependent variable corresponds to the log-odds ratio $\lambda_{ip,t} = \log(y_{ip}/(1 - y_{ip}))$ and can be written as:

$$\lambda_{ip,t} = \beta_0 + \beta_1 \text{sex}_i + \beta_2 \text{metoo}_p + \beta_3 \text{covid}_p + \beta_{12} \text{sex}_i \times \text{metoo}_p + \beta_{13} \text{sex}_i \times \text{covid}_p + \beta_4 X_{ip,t} + \varepsilon_{ip,t}. \quad (2)$$

The β coefficients thus capture the change in the proportion y_{ip} for a unit change in the predictors, which we define as follows: sex_i is an indicator that author i is a woman, metoo_p indicates that paper p was published post-MeToo, and covid_p indicates that paper p was published after the start of the pandemic.¹³ The control variables $X_{ip,t}$ include: (i) the period of publication, (ii) the characteristics of author i (e.g. professional experience and experience squared, research productivity, author degree and clustering up to the period prior to the publication of paper p), (iii) characteristics of paper p (e.g. number of authors, and field), and (iv) the relevant yearly average share, $r(y_{ip}, t)$, as described in section 2.2., to account for potential composition changes in the pool of authors over time that could confound our estimates. We cluster the standard errors at the author level to allow for within author correlation.

Notably, in non-linear models like ours, the interaction effect of metoo_p and sex_i on y_{ip} is more complex than in linear models. Denoting the vector of covariates in (2) with $\mathbf{X}$ and with $\beta = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_{12}, \beta_{13}, \beta_4)^T$, the interaction effect of metoo_p and sex_i on y_{ip} is given by¹⁴

$$\Delta^2 \mathbb{E}(y_{ip} | \mathbf{X}) / (\Delta \text{metoo}_p \Delta \text{sex}_i) = \Delta^2 \Lambda(\mathbf{X}, \hat{\beta}) / (\Delta \text{metoo}_p \Delta \text{sex}_i),$$

which is a function of covariates and all parameters in (2), making the effect of this interaction on the probability of collaboration dependent on the values of other covariates.^{15,16} To properly interpret these effects, we follow Karaca-Mandic et al. (2012) and compute average marginal effects, which show how the probability of the outcome changes with respect to a change in the variable of interest, holding other variables constant. We calculate the standard error of the marginal effects at the means of the explanatory variables using the delta method.

We use a logistic regression model to analyse our dependent variables, which are proportions bounded be-

¹³The covid_p dummy is interacted with the sex indicator, because the pandemic also had different effects on coauthorships patterns for men and women.

¹⁴We use the notation Δ to emphasize the discrete nature of metoo_p and sex_i , however this feature of non-linear models generalises to the case of continuous covariates, where

$$\frac{\partial^2 \mathbb{E}(y_{ip} | \mathbf{X})}{\partial x_2 \partial x_1} = \beta_{12} \Lambda'(\mathbf{X}\beta) + (\beta_1 + \beta_{12}x_2)(\beta_2 + \beta_{12}x_1) \Lambda''(\mathbf{X}\beta) \neq \beta_{12}.$$

¹⁵Moreover, as pointed out by Ai and Norton (2003), this has several implications. First, the interaction effect could be non-zero even if $\beta_{12} = 0$. Second, the statistical significance of the interaction effect cannot be tested with a simple t -test on the coefficient of the interaction term, β_{12} . Lastly, the interaction effect depends on all the parameters and the values of all covariates and may have different signs for different values of these covariates.

¹⁶In particular,

$$\begin{aligned} \frac{\Delta^2 \mathbb{E}(y_{ip} | \mathbf{X})}{\Delta \text{metoo}_p \Delta \text{sex}_i} &= \Lambda(\beta_0 + \beta_1 + \beta_2 + \beta_3 \text{covid}_p + \beta_{12} + \beta_{13} \text{covid}_p + \beta_4 z_{ip,t}) - \Lambda(\beta_0 + \beta_1 + \beta_3 \text{covid}_p + \beta_{13} \text{covid}_p + \beta_4 z_{ip,t}) \\ &\quad - \Lambda(\beta_0 + \beta_2 + \beta_3 \text{covid}_p + \beta_4 z_{ip,t}) + \Lambda(\beta_0 + \beta_3 \text{covid}_p + \beta_4 z_{ip,t}). \end{aligned}$$

tween 0 and 1. This approach allows for a flexible modelling of the response variable while ensuring predicted values remain within the unit interval (Papke and Wooldridge, 1996). We estimate the model using Maximum Likelihood Estimation (MLE), to obtain efficient and unbiased parameter estimates under correct specification. To avoid oversampling papers with larger numbers of authors, we weight observations by the inverse of the number of in-sample authors in each paper, following the approach of Solon, Haider and Wooldridge (2015).¹⁷

The nature of the MeToo movement and our data present challenges in establishing strict causal relationships. However, in Section 4, we use several strategies to rule out alternative explanations and argue that our specification isolates post-MeToo changes in coauthorship patterns that go beyond what can be attributed to general temporal trends, variations in the gender and seniority composition of authors over time. We quantify these effects as changes in coauthorship shares post-MeToo, based on our estimated marginal effects that measure the change in the predicted probability of the outcome after the MeToo period begins.¹⁸

3.1 Main Results

We estimate equation (2) for each of our dependent variables, which represent shares of coauthors by seniority and gender.¹⁹ Our primary focus is on the coefficients β_2 and β_{12} , which capture the expected change in the proportion of specific types of coauthors when the MeToo indicator switches on, allowing for differential effects by author gender. We estimate these models for the full sample and separately for the subsamples of junior and senior authors (see Section 2.5 for subsample descriptions).

Tables 2 and 3 present the estimated marginal effects of the MeToo indicator and its interaction with gender for all three samples. Each table contains three panels corresponding to our subsamples: all authors, junior authors, and senior authors. Within each panel, the first row (titled *All*) shows the estimated marginal effects from a restricted version of equation (2) that omits gender interactions (i.e., setting $\beta_{12} = \beta_{13} = 0$). The subsequent rows, titled *Men* and *Women*, present the gender-specific marginal effects from the unrestricted equation (2). Figure 4 visually presents selected estimated marginal effects.

Three main results stand out from our analysis. First, we observe an increase in mixed-gender collaborations within authors' seniority groups. As illustrated in Figure 4, Panel (a), the proportion of women coauthors for men has risen, both overall and within senior and junior subgroups.²⁰ Conversely, we find a decrease in the proportion of women coauthors for women authors, both overall and within corresponding seniority levels, although this effect is weaker for the junior subsample.²¹

Second, we observe a decline in collaborations between senior and junior economists. As illustrated in Figure 4, Panel (b), we estimate statistically significant decreases in the proportion of senior coauthors (especially senior women) for junior authors, and symmetrically, in the proportion of junior coauthors (particularly junior women) for senior authors.²² This finding, in combination with the increase in mixed-gender collaborations,

¹⁷Our data structure is a cross-section of author-paper combinations, not a panel.

¹⁸In logistic regression, the marginal effect of MeToo on y_{ip} is not constant and depends on the values of other covariates. We compute average marginal effects (AMEs) to summarise the impact of MeToo. For models with interactions (e.g., between MeToo and author sex), we calculate conditional marginal effects to capture how the MeToo effect varies across subgroups. For a detailed discussion on interpreting interaction effects in non-linear models, see Ai and Norton (2003).

¹⁹Specifically, y_{ip} represents the share of coauthors and new coauthors who are: senior, midcareer, junior, early-career, women, senior women, midcareer women, junior women, or early-career women.

²⁰These estimates correspond to the second and third rows of column (5) in the first panel, the second and third rows of column (8) in the second panel, and second and third rows of column (6) in the third panel in Table 2.

²¹A competing explanation for the changes we see in our analysis could be given by the circulation and popularisation of the working paper by Sarsons, Gërkhani, Reuben and Schram in 2017 (see Sarsons et al. (2021)). The paper suggests that credit attribution of collaborations in academic work is higher for men than for women, which would presumably decrease the incentives of women to collaborate with men. The increase in the proportion cross-gender collaborations goes against the credit-attribution story.

²²These estimates correspond to the first row of columns (1) and (6) in the second panel, and the first row of columns (3) and (8) in

Table 2: Marginal effects of MeToo on proportions of all coauthors in papers from CEPR and NBER.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Men and Women			Women					
	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 43,860$									
All	-0.0190** (0.0092)	0.0166** (0.0082)	-0.0089 (0.0078)	0.0161*** (0.0053)	0.0033 (0.0062)	-0.0035 (0.0041)	0.0057 (0.0038)	-0.0034 (0.0047)	0.0055 (0.0033)
Men	-0.0213** (0.0094)	0.0208** (0.0087)	-0.0076 (0.0081)	0.0145*** (0.0054)	0.0127** (0.0055)	0.0044* (0.0025)	0.0078*** (0.0027)	-0.0001 (0.0036)	0.0052* (0.0031)
Women	-0.0086 (0.0147)	-0.0000 (0.0122)	-0.0141 (0.0120)	0.0231*** (0.0081)	-0.0445*** (0.0158)	-0.0446** (0.0175)	-0.0012 (0.0124)	-0.0180 (0.0132)	0.0069 (0.0060)
Junior authors, $N = 7,787$									
All	-0.0189 (0.0171)	-0.0033 (0.0124)	0.0159 (0.0141)	0.0094 (0.0083)	-0.0052 (0.0137)	-0.0261*** (0.0077)	0.0043 (0.0062)	0.0122 (0.0098)	-0.0014 (0.0043)
Men	-0.0186 (0.0177)	-0.0057 (0.0128)	0.0208 (0.0147)	0.0068 (0.0084)	0.0132 (0.0110)	-0.0203*** (0.0075)	0.0049 (0.0060)	0.0174*** (0.0057)	-0.0017 (0.0038)
Women	-0.0200 (0.0228)	0.0046 (0.0168)	-0.0011 (0.0183)	0.0178 (0.0117)	-0.0754*** (0.0287)	-0.0446*** (0.0114)	0.0025 (0.0094)	-0.0072 (0.0258)	-0.0004 (0.0074)
Senior authors, $N = 26,287$									
All	-0.0134 (0.0084)	0.0039 (0.0074)	-0.0163** (0.0080)	0.0194*** (0.0066)	-0.0035 (0.0075)	-0.0013 (0.0041)	0.0029 (0.0043)	-0.0130*** (0.0043)	0.0082** (0.0042)
Men	-0.0121 (0.0086)	0.0023 (0.0075)	-0.0153* (0.0083)	0.0188*** (0.0068)	0.0059 (0.0070)	0.0067** (0.0027)	0.0028 (0.0042)	-0.0115*** (0.0044)	0.0083** (0.0041)
Women	-0.0207 (0.0136)	0.0123 (0.0113)	-0.0217* (0.0121)	0.0229** (0.0105)	-0.0659*** (0.0208)	-0.0483** (0.0200)	0.0038 (0.0082)	-0.0218*** (0.0072)	0.0082 (0.0070)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the nine outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

suggests a shift in collaboration patterns: there is an increase in sorting by seniority and a decrease in sorting by gender. Importantly, the gains in collaboration across genders do not stem from coauthorships across different seniority groups. Instead, we observe that junior men and junior women are collaborating more with each other, and similarly, senior men and senior women are increasing their collaborations within their seniority group. This pattern indicates potential sorting by seniority, even as gender barriers in collaboration appear to be diminishing within these seniority groups.

A third important result suggests a general chilling effect on the expansion of economists’ professional networks post-MeToo. As illustrated in [Figure 4, Panel \(c\)](#), we estimate decreases in the share of new coauthors across all seniorities and genders. Specifically, we observe declines in the share of new senior coauthors for juniors and new junior coauthors for seniors.²³ Notably, in the senior sample, we find a particularly significant decrease in new coauthorships with junior women. This trend aligns with recent findings by [Gertsberg \(2024\)](#), who observed that female scholars initiated fewer projects with new male co-authors after MeToo, especially

the third panel in [Table 2](#).

²³These estimates correspond to the first row of column (1) in the first panel, the first row of column (2) in the second panel, and the first row of column (4) in the third panel in [Table 3](#).

Table 3: Marginal effects of MeToo on proportions of new coauthors in papers from CEPR and NBER.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	New Men and Women					New Women				
	All	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 43,860$										
All	-0.0182*** (0.0070)	-0.0164*** (0.0052)	-0.0071* (0.0041)	-0.0067 (0.0055)	0.0156*** (0.0050)	-0.0038 (0.0049)	-0.0051** (0.0021)	0.0007 (0.0021)	-0.0040 (0.0032)	0.0041 (0.0029)
Men	-0.0193*** (0.0072)	-0.0164*** (0.0053)	-0.0076* (0.0042)	-0.0055 (0.0057)	0.0142*** (0.0051)	-0.0009 (0.0048)	-0.0033 (0.0020)	0.0007 (0.0020)	-0.0026 (0.0032)	0.0040 (0.0029)
Women	-0.0131 (0.0102)	-0.0164** (0.0072)	-0.0050 (0.0057)	-0.0115 (0.0077)	0.0217*** (0.0074)	-0.0163** (0.0077)	-0.0131*** (0.0033)	0.0007 (0.0036)	-0.0102** (0.0051)	0.0043 (0.0046)
Junior authors, $N = 7,787$										
All	-0.0213 (0.0155)	-0.0298** (0.0131)	-0.0045 (0.0099)	0.0072 (0.0105)	0.0088 (0.0080)	-0.0137 (0.0096)	-0.0197*** (0.0067)	0.0004 (0.0045)	0.0041 (0.0063)	-0.0030 (0.0041)
Men	-0.0155 (0.0159)	-0.0293** (0.0135)	-0.0044 (0.0099)	0.0136 (0.0112)	0.0071 (0.0082)	-0.0045 (0.0094)	-0.0134** (0.0060)	0.0004 (0.0043)	0.0076 (0.0063)	-0.0024 (0.0038)
Women	-0.0409* (0.0217)	-0.0317* (0.0165)	-0.0047 (0.0139)	-0.0155 (0.0129)	0.0143 (0.0108)	-0.0424*** (0.0133)	-0.0442*** (0.0118)	0.0006 (0.0073)	-0.0071 (0.0081)	-0.0045 (0.0064)
Senior authors, $N = 26,287$										
All	-0.0217** (0.0095)	-0.0125** (0.0060)	-0.0075 (0.0055)	-0.0193*** (0.0067)	0.0186*** (0.0063)	-0.0082 (0.0056)	-0.0043* (0.0023)	0.0012 (0.0030)	-0.0120*** (0.0036)	0.0060* (0.0036)
Men	-0.0244** (0.0098)	-0.0128** (0.0062)	-0.0091 (0.0056)	-0.0193*** (0.0068)	0.0177*** (0.0064)	-0.0058 (0.0057)	-0.0029 (0.0023)	0.0011 (0.0029)	-0.0106*** (0.0035)	0.0060* (0.0035)
Women	-0.0063 (0.0146)	-0.0108 (0.0089)	0.0007 (0.0080)	-0.0191* (0.0099)	0.0235** (0.0097)	-0.0216** (0.0094)	-0.0120*** (0.0040)	0.0019 (0.0054)	-0.0195*** (0.0058)	0.0063 (0.0062)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the ten outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

within the same institution.

A particularly interesting finding is the increase in coauthorships and new collaborations with early-career economists across our sample. This trend is evidenced by the positive estimates in the first panel of column (4) in Table 2 and column (5) in Table 3. Notably, the second panel of these tables suggests that this effect is primarily driven by senior authors, who show increased shares of both overall and new coauthorships with early-career economists, regardless of gender. This pattern may reflect two emerging trends in economics: the growing practice of including PhD students as formal coauthors, and the more recent inclusion of pre-doctoral researchers in papers with larger teams of senior economists. These shifts in collaboration patterns with early-career researchers stand in contrast to the general decline in new collaborations observed across other seniority levels, highlighting a potential area of change in academic mentorship and co-authorship practices post-MeToo.

The results for midcareer authors exhibit greater variability in both direction and statistical significance compared to other seniority groups. Nevertheless, we see a consistent decline in new coauthorships with mid-career authors across all three samples (all authors, juniors, and seniors). While this decline reaches statistical significance only in the full sample of all authors, it aligns with the overall trend of reduced new collaborations post-MeToo. This suggests that although the effects on midcareer collaborations are less pronounced, they still reflect the broader pattern of decreased new coauthorship formations observed across other seniority levels.

Table 4: Predicted percentage changes of shares of coauthors in papers from CEPR and NBER.

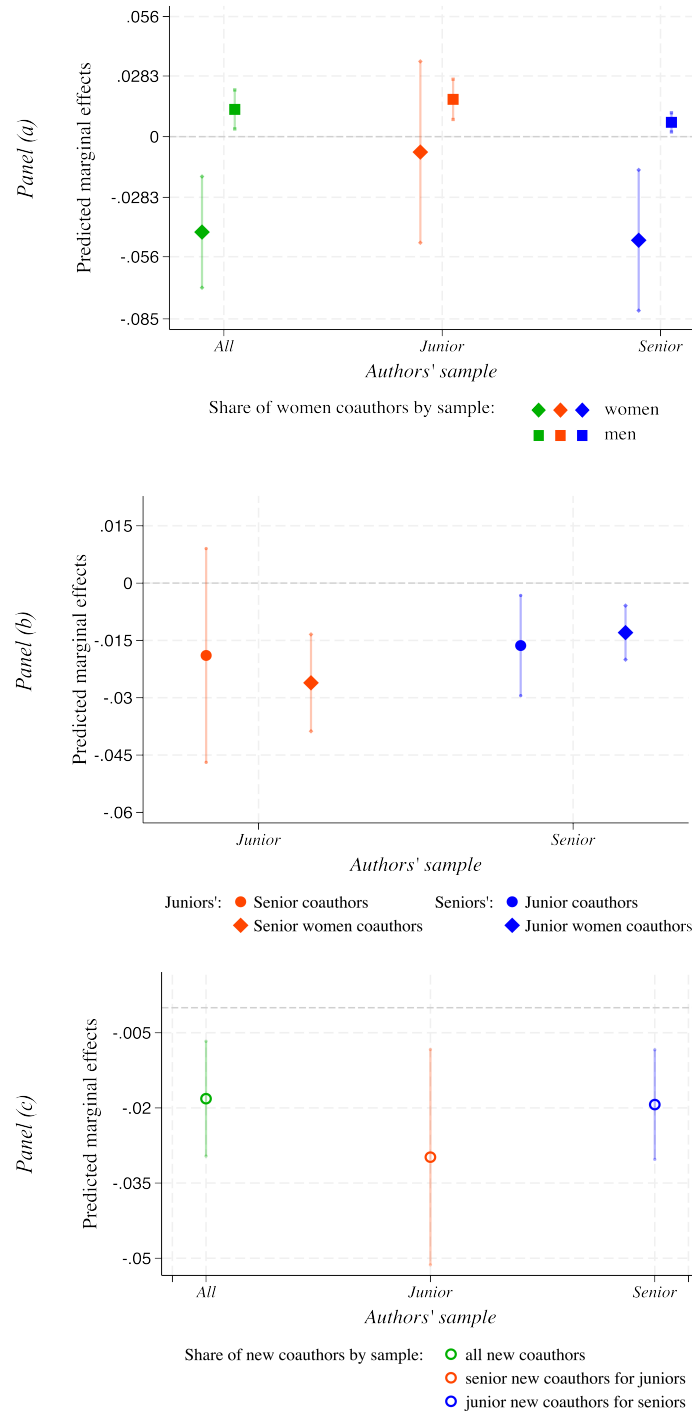
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
		Senior	Mid-career	Junior	Early-career	All women	Senior women	Mid-career women	Junior women	Early-career women
<i>All authors</i>										
All		-4.03	8.92	-3.74	24.39	1.66	-5.56	13.26	-5.40	28.95
Men		-4.40	11.30	-3.29	22.31	12.33	12.94	37.14	-0.34	43.33
Women		-2.01	0.00	-5.28	33.48	-8.05	-26.24	-0.94	-9.63	16.05
<i>Junior authors</i>										
All		-2.68	-3.40	13.14	18.80	-2.92	-26.10	20.48	36.97	-9.33
Men		-2.64	-5.88	17.05	13.60	12.11	-56.39	24.50	56.13	-12.14
Women		-2.83	4.95	-0.95	33.58	-13.39	-9.80	8.62	-15.65	-1.90
<i>Senior authors</i>										
All		-4.62	3.71	-2.96	52.43	-1.61	-3.71	12.08	-9.15	82.00
Men		-4.19	2.19	-2.77	52.22	6.28	23.10	14.00	-37.10	103.75
Women		-7.04	11.83	-4.00	58.72	-11.73	-91.13	10.56	-4.84	54.67
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	New Coauthors	New senior	New mid-career	New junior	New early-career	New women	New senior women	New mid-career women	New junior women	New early-career women
<i>All authors</i>										
All	-5.38	-12.24	-12.03	-7.44	42.16	-5.21	-26.84	5.00	-17.39	37.27
Men	-5.71	-12.33	-12.67	-6.11	37.37	-1.36	-20.63	5.38	-12.38	36.36
Women	-3.88	-11.88	-8.62	-12.92	62.00	-16.63	-46.79	3.89	-31.88	35.83
<i>Junior authors</i>										
All	-6.25	-21.75	-8.18	8.37	20.47	-17.79	-98.50	2.86	17.08	-23.08
Men	-4.53	-21.39	-7.72	15.81	16.51	-6.34	-74.44	3.08	34.55	-20.00
Women	-12.03	-22.48	-9.59	-18.67	31.09	-38.90	-138.13	3.53	-21.52	-25.00
<i>Senior authors</i>										
All	-6.52	-10.08	-12.71	-18.38	56.36	-11.71	-25.29	8.57	-48.00	66.67
Men	-7.42	-10.58	-15.69	-18.21	55.31	-9.83	-20.71	9.17	-50.48	75.00
Women	-1.83	-8.06	1.17	-18.73	67.14	-21.82	-48.00	10.00	-52.70	48.46

Notes: The top block shows predicted percentage changes of coauthorship shares, for the three samples *All authors*, *Junior authors* and *Senior authors*, by seniority of coauthors and gender. The bottom block shows predicted percentage changes of *new* coauthorship shares, for the three samples *All authors*, *Junior authors* and *Senior authors*, by seniority of coauthors and gender. All numbers are in % units. Predicted percentage changes are calculated using the baseline estimates from specification (2) and the corresponding pre-MeToo averages per 100 author-paper observations. Numbers in boldface are based on estimates that are significant at the 90% level.

To facilitate interpretation, we show the *percentage* changes in the corresponding shares of coauthorships relative to average shares pre-MeToo in [Table 4](#), where the panels and rows correspond to the ones we have in [Tables 2](#) and [Tables 3](#). The pre-MeToo averages are calculated using all author-paper observations up to July 1, 2018, our baseline start date for the MeToo period. The table includes results for each of our samples (all authors, juniors, and seniors). Values in boldface are statistically significant at the 90% confidence level and can be interpreted both as percent changes in coauthorship shares relative to pre-MeToo levels and as percent changes in the number of specific types of coauthors per 100 author-paper observations.

The results suggest that the estimated marginal effects are quantitatively sizeable. Using pre-MeToo average coauthors per paper, we can infer the average changes in the number of authors of a specific seniority or gender. For example, post-MeToo, for every 100 men-authored papers from NBER and CEPR, there are approximately 4 more women coauthors compared to the pre-MeToo average of 33 women per 100 men-authored papers. Conversely, for every 100 women-authored papers, there are about 11 fewer women involved post-MeToo. These numbers correspond to the estimated 12.3% increase in women coauthors for men authors and 8.1% decrease for women authors in the sample with all authors, as shown in column (6) of [Table 4](#). The post-MeToo divergence between senior and junior authors is apparent in columns (2) and (4): the share of junior authors' coauthorships with senior coauthors decreased by 2.7%, while the share of senior authors' coauthorships with junior coauthors dropped by 3.0%. These changes are driven by substantial declines in new coauthorships: a 21.8% decrease in junior authors' new coauthorships with senior coauthors, and an 18.4% reduction in senior authors' new coauthorships with junior coauthors. Notably, senior authors exhibit a striking 48% decline in the share of new coauthorships specifically with junior women, as shown in column (9).

Figure 4: Predicted Marginal Effects on Cross-gender Collaborations



Notes: The panels in the figure show the estimated marginal effects of the MeToo period indicator from equation (2) with corresponding 90% confidence bands. These effects capture the expected change in the proportion of specific types of coauthors when the MeToo indicator switches on, assuming all other factors remain constant. *Panel (a)* at the top displays estimates for the proportion of women coauthors. Green estimates correspond to the whole sample, orange to junior economists, and blue to senior economists. *Panel (b)* in the middle focuses on junior and senior economists. Orange estimates represent junior economists, while blue represents senior economists. *Panel (c)* at the bottom shows estimates for new coauthorships. Green estimates represent the share of new coauthors in the whole sample, orange represents new senior coauthors for junior economists, and blue represents new junior coauthors for senior economists.

4 Robustness and Validity of our Findings

In this section we address potential sources of bias. While we cannot categorically establish causality, we use several strategies to rule out alternative explanations we could be attributing to the MeToo movement rather than other concurrent trends or pre-existing patterns. Then we discuss the robustness of our findings in [Section 4.3](#). Tables and details of some of the results below are available in [Appendix C](#).

4.1 Placebo MeToo Date

First, to rule out the possibility that our results are capturing effects unrelated to the MeToo movement or are simply due to chance, we conduct a placebo test. We replace the original MeToo period with an earlier ‘placebo’ period of the same duration, from July 1, 2014, to December 31, 2016. This period was chosen as it was relatively uneventful in terms of major events affecting coauthorships and research productivity in economics. If our estimated effects are truly capturing the impact of the MeToo movement, we would expect to see different results for this unrelated period. As detailed in [Appendix C.1](#), we find small or no effects on coauthorship patterns during this placebo period. This stark contrast with our main findings strengthens the argument that the effects we observe are specifically related to the MeToo period rather than general trends or random fluctuations.

4.2 Time to Publication and the Timing of MeToo

Second, we try to leverage cross-sectional variation in addition to time variation to strengthen our argument that our findings are driven by the MeToo movement. To do so, we exploit the heterogeneity in the expected time it takes to write different types of working papers. While it is challenging to precisely determine the average time from inception to working paper draft, we know anecdotally that different types of research have different timelines. For instance, papers using structural models or field experiments typically take longer to complete than reduced-form analyses using secondary data.

To exploit this heterogeneity of the timelines in our data, we use Natural Language Processing (NLP) tools to classify papers based on their titles. We identify potentially ‘long-term projects’ by searching for phrases such as “field experiment”, “intervention”, “administrative”, or “quasi-experiment” in paper titles.²⁴ This approach allows us to construct a proxy distinguishing between slower and quicker papers, encompassing several hundred more papers than we could, if we were to rely solely on JEL codes.²⁵

To complement our analysis of longer-term projects, we also identified ‘faster’ papers in our dataset. We classified papers as potentially quicker to produce if they were theoretical in nature, based on specific JEL codes. These included category C (Mathematical and Quantitative Methods), excluding C9 (Design of Experiments), as well as selected D codes (D11, D21, D4, D82, D85, D86) related to theoretical aspects of microeconomics. This classification rests on the assumption that theoretical work typically has shorter production timelines compared to empirical research or field experiments. Unlike our classification of longer-term papers, we relied solely on JEL codes for this categorisation, as theoretical papers often lack clear indicators in their titles that would distinguish them as ‘faster’ projects. However, this approach has limitations. Notably,

²⁴Ideally, we would apply these techniques to the abstracts of papers rather than just their titles. However, abstracts are not available in our data.

²⁵For example, the only JEL code we could easily identify as indicating a slower paper is C96 indicating field experiments. However, this JEL code appears in our sample in less than 200 papers out of more than 26,700 distinct papers available. While we identify a similar number of papers relying on field experiments when we use titles, we can expand our criterion for determining whether a paper is slow by using additional terms by using NLP.

theoretical economics is a field that has been historically male-dominated, which complicates the interpretation of gender-related analyses when focusing on these ‘faster’ papers.

These classifications creates variation in how the MeToo movement might affect papers with different completion timelines, depending on their methodologies. This variation allows us to gain insights into whether the effects we observe are indeed attributable to the MeToo movement. If our findings are driven by MeToo, we would expect to see more immediate effects in quicker papers, while longer-term projects might show a delayed response. This differential impact provides an additional layer of identification, helping to distinguish MeToo effects from other concurrent trends or events that might affect all types of papers similarly.

We repeated our analysis first excluding slower papers and then excluding faster papers. When we exclude longer-term papers from our analysis, our main findings not only remain qualitatively similar but become stronger in some cases. The increase in mixed-gender collaborations, remains consistent but shows stronger effects when we focus on the decrease in women collaborating with other women. The decline in senior-junior co-authorship persists, with a notably stronger effect for senior authors collaborating with juniors. While the overall chilling effect on network expansion slightly weakens for junior authors, it strengthens significantly for senior authors. These slightly more pronounced effects when we exclude longer papers aligns our expectation that the MeToo movement’s impact would be more immediately visible when exclude slower-to-produce papers. The detailed results of this analysis are available in [Table C.3](#) and [Table C.4](#) in [Appendix C.2](#).

A significant challenge in this analysis is that papers that take extended periods to produce often undergo changes in their authorship team over time. Some coauthors may remain as symbolic collaborators due to their past contributions, while others may be added later in the project as it evolves. Therefore, a paper conceptualised years before the MeToo movement could potentially incorporate post-MeToo coauthors in its final stages. This dynamic nature of authorship creates a potential confounding factor that makes it challenging to cleanly separate pre- and post-MeToo collaboration decisions.

When we exclude quicker papers from our analysis, our main findings remain qualitatively consistent, but with notable variations in effect sizes across subsamples. The increase in mixed-gender collaborations persists, albeit with weaker effects for men overall. Interestingly, we observe a further decrease in women’s share of male coauthors. This pattern might suggest that men were more likely to swiftly adjust their collaboration patterns in response to MeToo in faster-turnaround papers, while women potentially adopted a more cautious approach to collaboration in longer-term projects. The decline in senior-junior co-authorship remains evident, with a stronger effect for junior authors collaborating with seniors, but a weaker one for senior authors collaborating with juniors. The overall chilling effect on network expansion intensifies for the full sample and for junior authors, but weakens for senior authors. While these heterogeneous changes do not fully align with our expectation that the MeToo movement’s impact would be less immediately visible in slower-turnaround papers, they do not invalidate this hypothesis either. Instead, they highlight the complex and nuanced nature of the MeToo movement’s influence on collaboration patterns across different types of research projects. Detailed results of this analysis are available in [Table C.5](#) and [Table C.6](#) in [Appendix C.2](#).

As we note above, relying only on JEL codes to define faster papers restricted out subsample to theoretical economics papers. Theoretical fields, however, are more male-dominated than most others, complicating the interpretation of gender-related analyses when focusing on these “faster” papers. Consequently, while this classification provides insight into potential differences in collaboration patterns for quicker-to-produce papers, the results should be interpreted with caution, particularly regarding gender dynamics.

4.3 Robustness Checks

Our findings are robust to estimating the OLS version of equation (2), where the dependent variables are the shares y_{ip} , and to reducing our explanatory variables to a minimal set of controls. We also estimated equation (2) expanding the cut-off for defining new coauthorships from 3 months to 0 and 6 months, whilst still removing the first time that an author appears with a paper in the sample. This change alters the number of coauthorships defined as *new* and only impacts results with shares of *new* coauthorships as dependent variables. The estimated marginal effects in both these two cases are very similar to those obtained with the 3-month cutoff, but we do observe that the larger the cutoff of months, the lower the estimated effects are.²⁶ These variations consistently support our main findings, with only minor variations in the magnitudes and statistical significance of the estimated effects, all in intuitively expected directions.

Having established the stability of our results under various specifications, we delve into validating our choice of the MeToo starting date and addressing potential gender-related factors that could confound our results.

4.3.1 Validating our choice of the MeToo date

A key challenge in our analysis is defining the post-MeToo era: the movement's gradual unfolding and the heterogeneity in the time it takes to complete a paper make a precise definition infeasible. To validate our choice for the start date of the MeToo period we conduct a sensitivity analysis on the starting date of the MeToo period.

We estimate our model multiple times, varying the beginning of the MeToo period from October 1, 2017, to December 1, 2019, in three-month increments. This approach yields nine estimates for the MeToo indicator switching on, corresponding to different starting dates of the MeToo dummy variable. This approach allows us to account for potential delayed impacts and additional effects associated with related events, such as the 'Professional Climate Survey' of the *American Economic Association*, which took place late in 2018, and the establishment of their 'Policy on Harassment and Discrimination' in early 2019. The results, presented graphically in Figure C.3, support our main findings. Figure C.3, Panel (a) shows the proportion of junior and new junior coauthors for the senior authors in our sample and Figure C.3, Panel (b) shows the proportion of junior and new junior women coauthors for the senior authors in our sample.

The only statistically significant effects correspond to MeToo periods starting in 2018. The effects are negative, implying declines in the corresponding coauthorship shares during the year of the aftermath of original MeToo in October 2017. These effects become statistically insignificant after the end of 2018, suggesting that the decline in the coauthor shares tapers off from that time onwards. Interestingly, the selected effects on new coauthorships, and in particular those of senior authors with juniors and junior women, are negative and statistically significant for most MeToo starting dates in 2018, suggesting a lasting and persistent effect of MeToo.

4.3.2 Accounting for Gender-Specific Effects

The academic experiences of men and women in economics, as in other sectors of the labour market, often differ systematically. For instance, women are more likely to take parental leave or experience career interruptions, potentially making our experience proxy less precise for them. These differences suggest that our control variables may have gender-specific effects on collaboration patterns. To account for these differences while

²⁶The results from all these variations are available from the authors upon request.

maintaining statistical power and allowing for direct comparability across genders, we estimated an extended version of our model. In this specification, we interacted all author-level characteristics with the sex indicator.

The results from this gender-interacted specification broadly support our main findings on post-MeToo collaboration patterns, with some notable variations (detailed results are available in [Appendix C.4](#)). The increase in mixed-gender collaborations, weakens for both genders but preserves significance. The decline in senior-junior coauthorships persists, with interesting gender differences. For juniors, the reduction in senior coauthors is weaker for men but stronger for women. Conversely, for seniors, the decline in junior coauthors is stronger for men and weaker for women. The general chilling effect on the expansion of economists' professional networks remains surprisingly similar in levels and significance.

These estimates reveal that variables such as network size, productivity, and experience indeed have differential effects on collaboration patterns for men and women. While the overall trends remain consistent with our baseline model, the gender-specific impacts suggest that factors like career interruptions play a role in shaping collaboration decisions. Although our experience proxy may not directly capture these interruptions, the observed differences in effects across genders indicate their importance in collaboration patterns. By allowing for these gender-specific effects, our analysis provides a more nuanced understanding of how the MeToo movement may have influenced collaboration patterns differently for men and women in economics. This approach strengthens our findings by demonstrating their robustness to gender-specific variations in key variables.

5 Heterogeneity across outlets and fields

5.1 Differences between the NBER and CEPR

As discussed in [Section 2.1](#), there are structural and institutional variations between the two research networks. The NBER is predominantly composed of economists based in the US, many of whom are affiliated with elite universities known for well-established hierarchies and mentoring structures, potentially resulting in a more tightly connected academic network. In contrast, even though mostly European-based, CEPR has a more internationally diverse membership, where institutional and academic cultures regarding collaboration, mentorship, and co-authorship may differ. These differences together with institutional and cultural differences in gender diversity policies between the U.S. and Europe could influence how senior and junior economists interact across the two networks.

Separately estimating (2) by outlet reveals heterogeneous effects across NBER and CEPR networks align to a large extent with our understanding of their structural and institutional differences. The estimates can be found in [Appendix D.1](#), [Tables D.1 to D.4](#).

The increase in mixed-gender collaborations, remains consistent with our baseline results but shows stronger effects when we restrict the sample to NBER publications. It remains present in the subsample of CEPR papers, but it largely loses statistical significance except for the case of senior women, who show a stronger reduction in collaborations with women in CEPR papers. These differences may reflect the more tightly connected nature of the US-based academic network and potentially more pronounced impact of the MeToo movement in this context. The CEPR's more diverse international membership might partly explain the weaker and often insignificant effects, since any impact of MeToo on collaboration patterns may be moderated by varied institutional cultures and policies across different countries.

Separately analysing the NBER and CEPR suggests that the decline in senior-junior coauthorship we find in our baseline sample is primarily driven by the NBER sample, where junior economists coauthor significantly

less with senior authors. In the CEPR sample, we find similar but weaker and statistically insignificant effects, for the most part. Notably, senior economists in the CEPR show a stronger reduction in collaborations with junior women. While the overall chilling effect on network expansion remains present across the two outlets, the decrease is stronger for junior collaborations with new seniors in the NBER dataset and stronger when considering all authors collectively in the CEPR than in our baseline results. These differences may reflect distinct approaches to mentorship and collaboration in European versus U.S. academic contexts.

To further corroborate that the U.S./Europe divide could partially explain these results, we also estimated Equation (2) separately by country of publication, using the author’s institution at the time of publication to determine the appropriate country. Despite a slight decrease in sample size for the U.S. (from 30,718 in the NBER sample to 27,754) and for Europe (from 18,631 in the CEPR sample to 12,742), the results remain qualitatively consistent and maintain similar significance levels. The effects are slightly weaker due to reduced sample sizes, but the overall patterns persist, lending further support to our interpretation of the NBER/CEPR differences.

These findings, summarised in Figure 5, suggest that while the MeToo movement has had a broad influence on economics research networks, its effects are not uniform and are shaped by the specific characteristics and structures of different academic communities.

5.2 Heterogeneity across Research Fields

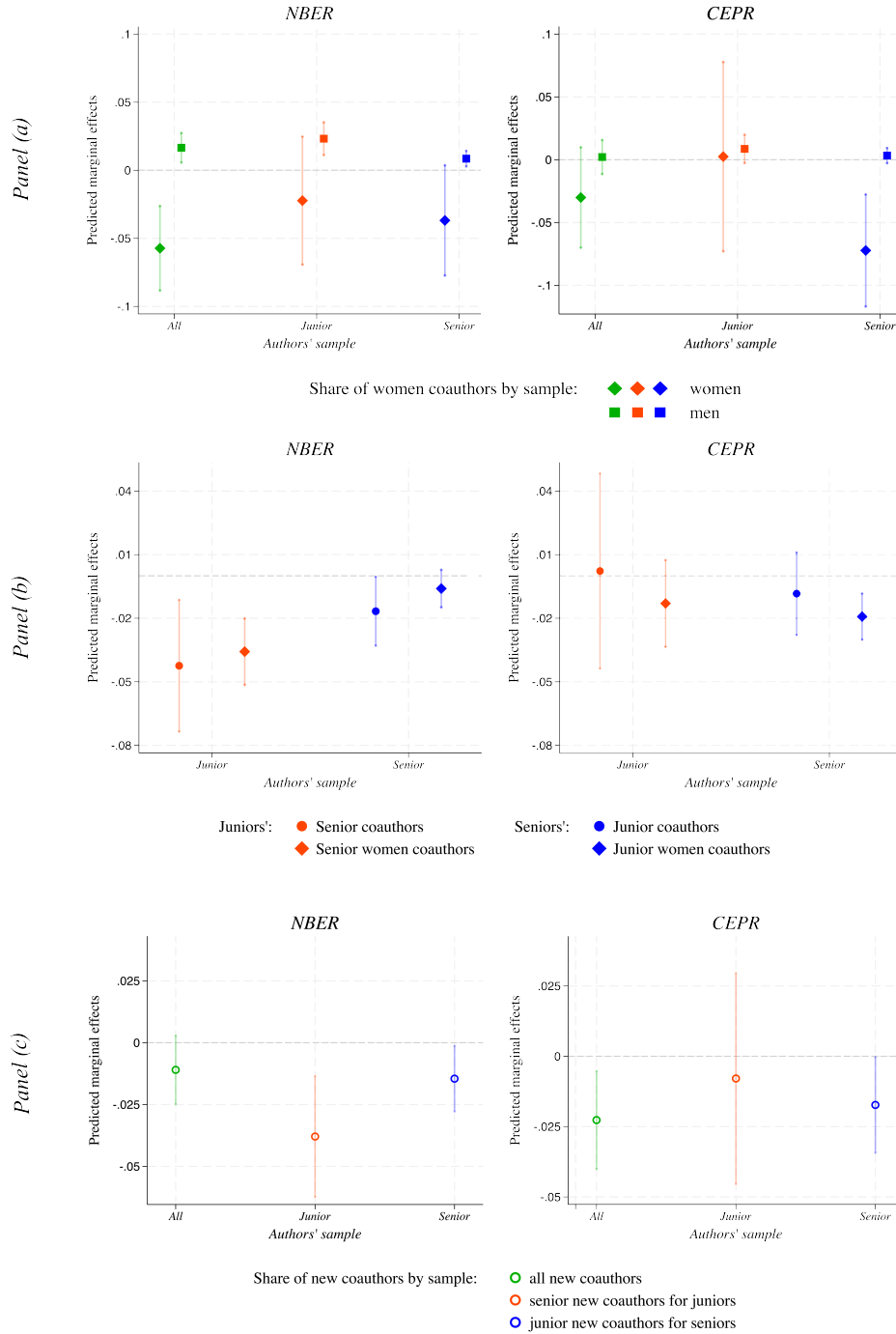
While our main findings show consistent patterns across various specifications and subsamples, the lack of significance in some of our estimates suggests the possibility of heterogeneous effects across different groups of economists. To explore this further, we conduct a field-specific analysis. This approach allows us to shed light on the types of heterogeneity across research areas that may be behind some of the insignificant effects observed in our aggregate results, providing a more nuanced understanding of how the MeToo movement has influenced collaboration patterns in different subfields of economics.

We categorised papers based on their Journal of Economic Literature (JEL) codes. Specifically, we grouped first-digit JEL codes into three main categories: (i) *applied micro*, including JEL codes H, I, J, L, N, O, and P, (ii) *macro* including JEL codes E, F and G, and (iii) *theory*, including JEL codes C and D.²⁷ For each paper, we calculated the proportion of its first-digit JEL codes falling into each group, and classified a paper as belonging to a particular group if the majority of its first-digit JEL codes were from that group. Authors were then assigned to groups based on the modal classification of their papers. Importantly, authors can belong to more than one group if their papers were evenly distributed across categories. This approach allows for a nuanced categorisation of research areas while accounting for the multidisciplinary nature of many economics papers.

Our field-specific analysis reveals significant heterogeneity in collaboration patterns across research areas. Figure D.1, Panel (a) reveals that the applied micro group appears to be driving the decrease in the mixed-gender collaborations for junior authors, while the macro fields largely influence gender-mixing for senior authors. Although the full sample shows a noisy decrease in junior women’s collaborations with other junior women (see Figure 4, Panel (a)), field-level analysis reveals heterogeneity across fields (see Figure D.1, Panel (a)). Specifically, macro and theory related fields show an increase, rather than decrease, in the share of junior women coauthoring with other junior women, though this effect is insignificant for the macro group and the theory group is heavily male-dominated. Figure D.1, Panel (b) illustrates that senior authors in macro-related fields exhibit a stronger decrease in their coauthorships with junior women compared to what we see in Figure

²⁷We also considered alternative groupings; for instance, excluding L and P from applied or including them in theory. These alternatives yielded similar but noisier results.

Figure 5: Predicted Marginal Effects on Cross-gender Collaborations by Publication Outlet



Notes: The panels in the figure show the estimated marginal effects of the MeToo period indicator from equation (2) with corresponding 90% confidence bands. These effects capture the expected change in the proportion of specific types of coauthors when the MeToo indicator switches on, assuming all other factors remain constant. *Panel (a)* at the top displays estimates for the proportion of women coauthors. Green estimates correspond to the whole sample, orange to junior economists, and blue to senior economists. *Panel (b)* in the middle focuses on junior and senior economists. Orange estimates represent junior economists, while blue represents senior economists. *Panel (c)* at the bottom shows estimates for new coauthorships. Green estimates represent the share of new coauthors in the whole sample, orange represents new senior coauthors for junior economists, and blue represents new junior coauthors for senior economists.

4, *Panel (b)*. Additionally, [Figure D.1, Panel \(c\)](#) suggests they form slightly fewer new collaborations with junior authors in general compared to our baseline in [Figure 4, Panel \(c\)](#).

These findings underscore important field-level differences in coauthorship patterns post-MeToo. They suggest that the impact of the MeToo movement on collaboration in economics may vary significantly across research areas, potentially due to differences in field-specific cultures, gender representation, or mentorship practices. This heterogeneity not only helps explain some of the insignificant effects in our aggregate results but also highlights the complexity of the MeToo movement’s influence on academic collaborations in economics.

6 Conclusion

Our analysis reveals significant shifts in the collaborative landscape of economics following the MeToo movement. These changes cannot be fully explained by general temporal trends, variations in the gender and seniority composition of authors over time, or other observable confounding factors in our data. Most notably, we document a surge in mixed-gender collaborations within seniority groups. This result is good news for the economics profession. If taken as causal, it could suggest that authors may have recognised that more diverse collaborations could be conducive to higher quality and more impactful research, as established by the literature ([Anderson and Richards-Shubik, 2022](#); [Maddi and Gingras, 2021](#)).

However, we also unveil an increase in collaborations within seniority groups and a notable decline in the initiation of new collaborations outside existing networks. This pattern is particularly pronounced among senior scholars, who have initiated fewer new coauthorships across all seniority levels, except with early-career economists. The substantial decrease in new collaborations between senior authors and junior researchers, with this trend being particularly pronounced for collaborations with junior women. This is bad news. If taken as causal, these results suggest two plausible interpretations: First, authors may have prioritised increasing gender diversity in their collaborations. Second, senior authors have increasingly relied on their existing collaboration networks rather than forming new coauthorships. The latter trend, if persistent, could have long-lasting consequences for the career development of women economists and potentially exacerbate the already ‘leaky’ pipeline in the profession.

Our findings also reveal interesting heterogeneous trends across different field groups and countries. For instance, macro and theory fields show distinct patterns in gender-mixing collaborations compared to applied micro fields. Future research should explore the mechanisms driving these changes and their implications for diversity and inclusivity in economics. Specifically, how these shifts affect mentorship, knowledge transfer, and career progression in the field, as well as explore strategies to promote more inclusive and diverse collaboration networks.

These results underscore the urgent need for sustained efforts to cultivate a supportive ecosystem and dismantle systemic barriers hindering the advancement of women and junior economists in the field. The economics profession must proactively continue to foster a safe, inclusive environment by evaluating, monitoring, and educating on relevant issues.

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Appendix A. Definitions of variables

A.1 Author Degree and Clustering

We define a variable of coauthorship for author i in paper p in period t with author j as

$$ca_{ip,t}(j) = \begin{cases} 1 & \text{if author } j \text{ is an author in paper } p \text{ written in period } t, \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

By definition, $ca_{ip}(i) = 1$.²⁸ The link of an author i with author j in period t is defined as

$$\alpha_{ij,t} = \begin{cases} 1 & \text{if } ca_{ip,t}(j) \neq 0 \text{ for at least one } p \text{ up to and including period } t, \\ 0 & \text{otherwise.} \end{cases}$$

We note that is a cumulative/stock variable, i.e. $\alpha_{ij,t'} \geq \alpha_{ij,t}$ for any $t' \geq t$. In words, once two authors have collaborated in a paper, they are considered coauthors thereafter. Based on this definition of author links, we define the *degree* for author i at time t as

$$d_{i,t} = \sum_{j=1}^M \alpha_{ij,t},$$

where M is the number of distinct authors in our sample. The *clustering* of author i at time t is

$$c_{i,t} = \frac{\sum_{i \neq j, k \neq j, i \neq k} \alpha_{ij,t} \alpha_{kj,t} \alpha_{ik,t}}{\sum_{i \neq j, k \neq j, i \neq k} \alpha_{ij,t} \alpha_{ik,t}}.$$

The clustering coefficient of an author i quantifies the likelihood that two distinct coauthors of author i have collaborated on a paper together. We note that given the nature of our dataset, this variable may underrepresent the clustering because pairs of such coauthors may have joint papers that do not appear in NBER or CEPR.

A.2 Example of coauthorship

To illustrate the definition of coauthorship, we present a simple example of a made-up sample of a total of four papers and three authors. We denote the papers as P1, P2, P3, and P4, with the numbering indicating chronological order (i.e., P1 appears first in the sample, and P4 appears last). We denote the authors as A, B, and C. Table A.1 below describes the data, where an ‘x’ indicates that the author (indicated in the rows) contributed to the paper (indicated in the columns). In this example, P1 is authored by A and B, P2 by A and

Table A.1: Example with three authors

	P1	P2	P3	P4
A	x	x		x
B	x		x	x
C		x	x	x

Notes: The table describes the toy data in the example, where an ‘x’ indicates that the author (indicated in the rows) contributed to the paper (indicated in the columns).

C, P3 by B and C, and P4 by all three authors (A, B, and C). For each author $j \in \{A, B, C\}$, equation (3)

²⁸Strictly speaking the subscript t is not needed for the definition of coauthorship but it is useful for defining the author variables *degree* and *clustering*.

defines coauthorships for author i in paper p with author j as shown in Table A.2, where we have marked with boldface the new coauthorships.

Table A.2: Coauthorships in example with three coauthors

Panel 1: A's coauthorships					Panel 2: B's coauthorships				
$ca_{Ap}(j)$	P1	P2	P3	P4	$ca_{Bp}(j)$	P1	P2	P3	P4
A	1	1	0	1	A	1	0	0	1
B	1	0	0	1	B	1	0	1	1
C	0	1	0	1	C	0	0	1	1

Panel 3: C's coauthorships				
$ca_{Cp}(j)$	P1	P2	P3	P4
A	0	1	0	1
B	0	0	1	1
C	0	1	1	1

Notes: The table displays the values of variable $ca_{ip}(j)$ in the toy data in our example. As an example, consider author B, in the second panel. Each row in this panel represents a potential coauthor (including B herself), and each column represents a paper. A '1' in a cell indicates that B coauthored with that person on that paper, while a '0' indicates no coauthorship. The first time a coauthorship forms, is marked in red.

As an example, consider author B, in the second panel of Table A.2. B is a coauthor of A and B herself in P1. She is not coauthor of C in P1. She is not a coauthor of anyone in P2 (indicated by 0's in the P2 column). She is a coauthor of herself and of C in P3, and a coauthor of A, C and herself in P4. The first time a coauthorship forms, is marked in red. Once a person is coauthor of B for the first time, all *future* coauthorships are considered not new. That is, since B is a coauthor of A in P1, their coauthorship in P4 which occurs later than P1, is considered old. Based on this reasoning, author B's new coauthorships are with A in P1, with B (herself) in P1, and with C in P3.

We can extend this example to include an additional author, author D, who participates only in paper P4. The tables illustrating this new setting that includes author D are Table A.3 and A.4—which are augmented versions of Table A.1 and A.2, respectively. Panel 4 in Table A.4 shows that D is not a coauthor of neither A, B or C in P1, P2, or P3 but is a coauthor of all three authors in P4. All these are new coauthorships for D in P4, and therefore marked with red. D is also a new coauthor for A, B and C, and is marked with red in their corresponding panels.

Table A.3: Example with four authors

	P1	P2	P3	P4
A	x	x		x
B	x		x	x
C		x	x	x
D				x

Notes: The table shows an augmented version of our original example in Table A.1, now including author D who participates only in paper P4.

To calculate the number (and shares) of new coauthorships for an author i in a given paper p , we add up the red 1's for the corresponding entry in each panel. For example, to calculate the number of new coauthorships for author i in paper $p = P1$, we look at the entry corresponding to P1 for each author in their respective panels.

Since new coauthorships include oneself, the number of new coauthorships for author A in P1 is 2 (out of the total of two coauthorships). The share of new coauthorships in this case is 100%. Similarly, to calculate the number of new coauthorships for C in P3, we add up red 1's in elements (3,3) from each grid: these are 1 (out of the total of two coauthorships C has). The share of new coauthorships is now 50%. Finally, to calculate new coauthorships for author B in P4, we add up the red 1's in entries (2,4) in each grid, i.e., 1 (out of the total of four coauthorships B has). In this case the share of new coauthorships is 25%.

In our regressions and analysis, we drop authors who appear in the sample in only one paper. However, we do this only after calculating all shares of coauthorships to ensure we capture the full network effects. This approach allows us to maintain a more stable sample of authors while still accounting for all collaboration information.

In our augmented example, we would thus remove author D from the sample, as she is only an author in P4. It's important to note that even after removing D, the share of new coauthorships for author B in P4 is still recorded as 25%. This preserves the impact of D's presence on the collaboration patterns of the remaining authors. However, we do not include D's own share of new coauthorships (which would be 100%) in our final analysis, as we remove this author from the sample.

This approach ensures that we capture the full extent of collaboration patterns while focusing our analysis on authors with more established presence in the network.

Table A.4: Coauthorships in example with four coauthors

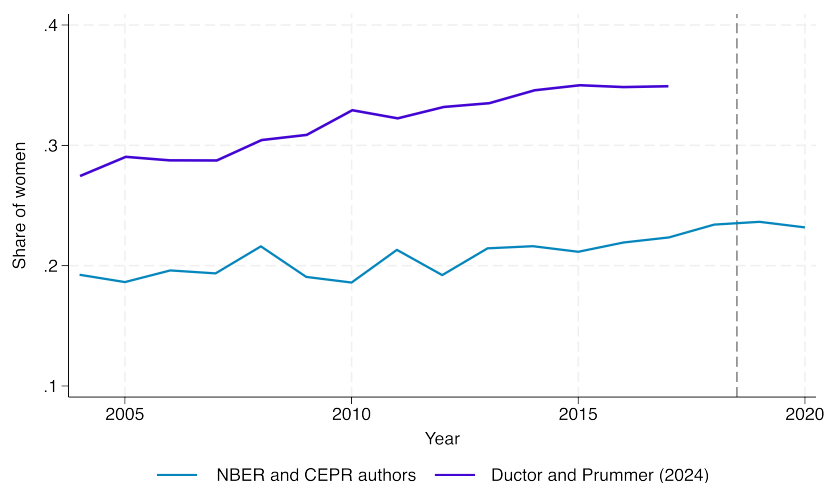
<i>Panel 1: A's coauthorships</i>					<i>Panel 2: B's coauthorships</i>				
$ca_{Ap}(j)$	P1	P2	P3	P4	$ca_{Bp}(j)$	P1	P2	P3	P4
A	1	1	0	1	A	1	0	0	1
B	1	0	0	1	B	1	0	1	1
C	0	1	0	1	C	0	0	1	1
D	0	0	0	1	D	0	0	0	1

<i>Panel 3: C's coauthorships</i>					<i>Panel 4: D's coauthorships</i>				
$ca_{Cp}(j)$	P1	P2	P3	P4	$ca_{Dp}(j)$	P1	P2	P3	P4
A	0	1	0	1	A	0	0	0	1
B	0	0	1	1	B	0	0	0	1
C	0	1	1	1	C	0	0	0	1
D	0	0	0	1	D	0	0	0	1

Notes: The table displays the values of variable $ca_{ip}(j)$ in the toy data in our example, including author D. For example, consider author C, in the third panel. Each row in this panel represents a potential coauthor (including C herself), and each column represents a paper. A '1' in a cell indicates that C coauthored with that person on that paper, while a '0' indicates no coauthorship. The first time a coauthorship forms, is marked in bold.

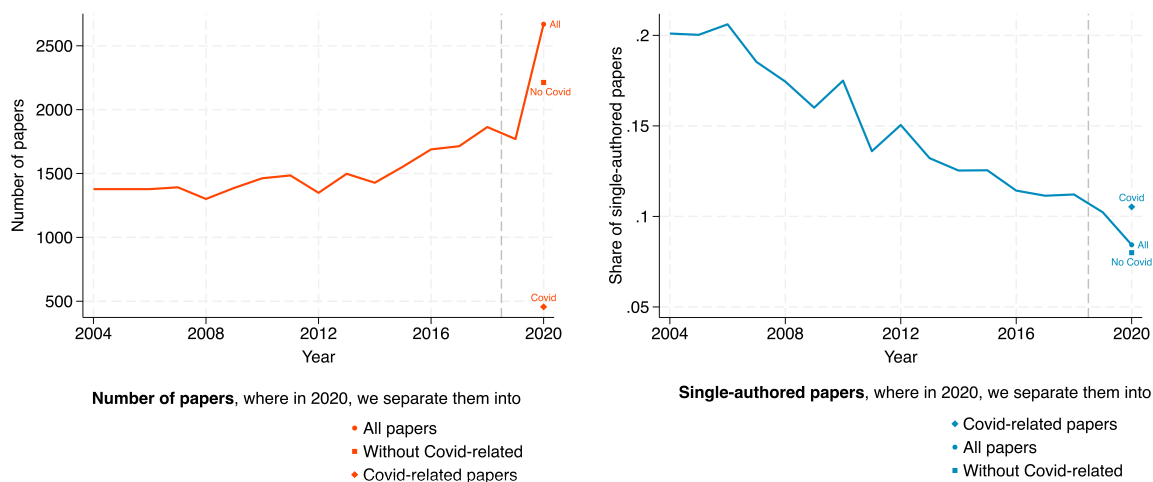
Appendix B. Data Trends and Summary Statistics

Figure B.1: Share of women in NBER & CEPR sample, vs sample of *Ductor and Prummer (2024)*



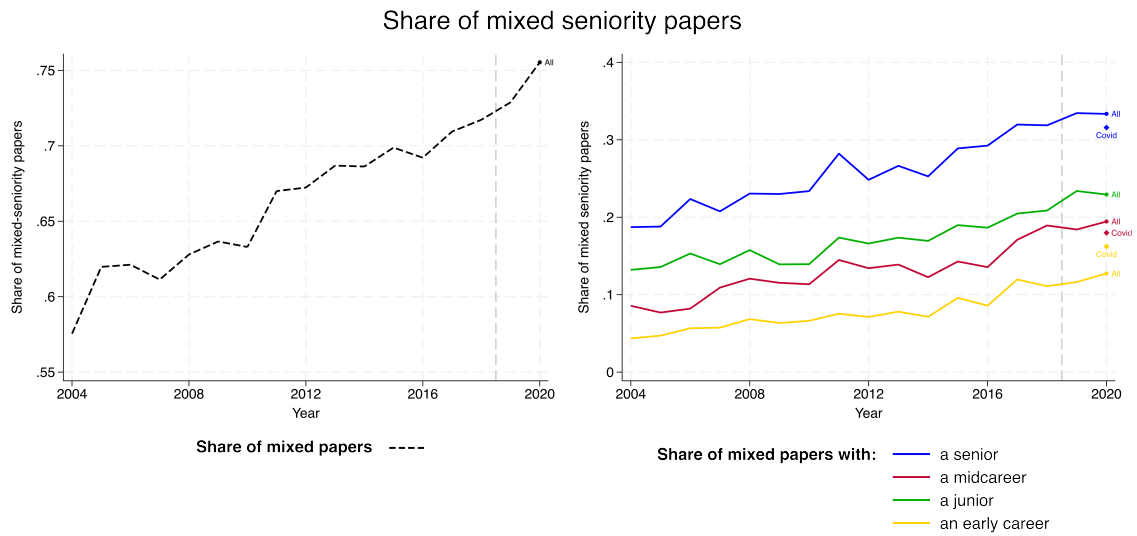
Notes: The figure shows the share of women authors that appear in the NBER or CEPR from 2004–2020 in blue. The purple line shows the share of women authors in the sample of *Ductor and Prummer (2024)*.

Figure B.2: Number of Papers and Share of Single-Authored per Year



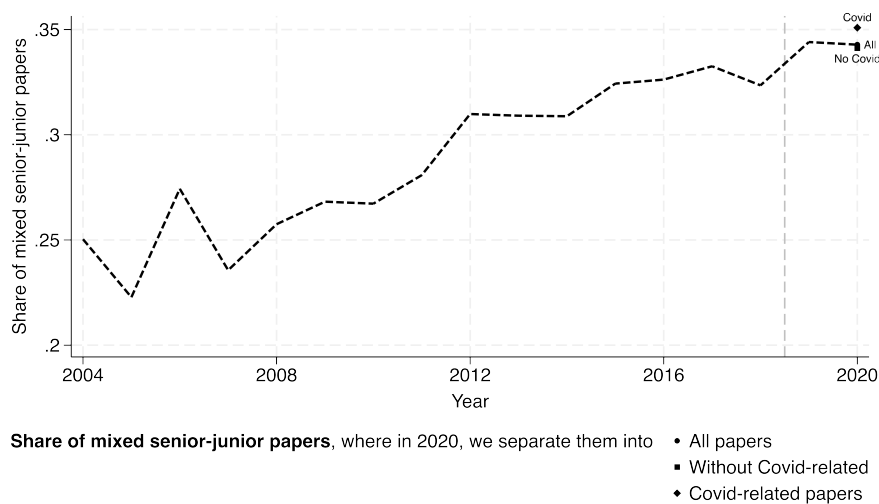
Notes: The figure shows on the left panel, the total number of papers published year by year in either NBER or CEPR from 2004–2020. The increasing trend spikes in 2020. This spike remains even if we do not consider Covid-related papers as can be seen by the square symbol, corresponding to 2,213 papers, a 25.03% increase compared to year to year changes ranging between -7 and 11%. The overall increase (including Covid-related papers) was 50%. The right panel shows the share of single-authored papers published year by year in either NBER or CEPR from 2004–2020. The share remained virtually the same from 2019 to 2020 if we consider Covid papers only, but it continued to decrease overall.

Figure B.3: Share of Mixed-Seniority Papers by Year



Notes: The figure shows, on the left panel, the share of mixed-seniority papers published year by year in either NBER or CEPR from 2004–2020. On the right panel it shows the shares of mixed papers with at least a member of the different seniority groups we considered. We do not label the differences between Covid-related or unrelated papers when the difference is negligible.

Figure B.4: Share of Mixed Senior-Junior Papers by Year



Notes: The figure shows the share of mixed-seniority papers that include at least one senior and one junior authors published year by year in either NBER or CEPR from 2004–2020.

Table B.1: Summary Statistics pre MeToo

	All	Men	Women
All authors			
Number of papers	18988		
Number of authors	6448	5063	1385
Authors' average characteristics			
Lagged degree	4.17	4.32	3.64
Lagged clustering	0.85	0.84	0.89
Years of experience	12.15	12.86	9.56
Number of coauthors per paper	2.90	2.90	2.91
Number of papers per period	2.04	2.10	1.83
Cumulative num. of papers	4.98	5.22	4.13
Authors' seniorities at publication			
Early career economists	367	255	102
Junior economists	1794	1310	484
Micareer economists	1365	1028	337
Senior economists	2693	2294	399
Unclassified	239	176	63
Junior authors			
Number of papers	5617		
Number of authors	2421	1780	641
Authors' average characteristics			
Lagged degree	2.98	3.06	2.75
Lagged clustering	0.92	0.92	0.93
Years of experience	5.09	5.03	5.25
Number of coauthors per paper	2.81	2.81	2.82
Number of papers per period	2.03	2.08	1.90
Cumulative num. of papers	3.66	3.76	3.39
Authors' seniorities at publication			
Early career economists	0	0	0
Junior economists	2421	1780	641
Micareer economists	0	0	0
Senior economists	0	0	0
Unclassified	0	0	0
Senior authors			
Number of papers	14910		
Number of authors	3025	2562	463
Authors' average characteristics			
Lagged degree	6.16	6.17	6.09
Lagged clustering	0.75	0.75	0.76
Years of experience	19.34	19.76	16.98
Number of coauthors per paper	2.86	2.86	2.91
Number of papers per period	2.33	2.38	2.12
Cumulative num. of papers	7.59	7.69	7.02
Authors' seniorities at publication			
Early career economists	0	0	0
Junior economists	0	0	0
Micareer economists	0	0	0
Senior economists	3025	2562	463
Unclassified	0	0	0

Notes: The pre-MeToo period corresponds to our full sample period up to 1 Jul 2018. The number of authors by seniority in *All Authors* sample is counted using the mode seniority of each distinct author. Average characteristics are calculated as the average across the mean characteristic of distinct authors.

Appendix C. Robustness and Validity of our Findings

C.1 Placebo MeToo Date

Table C.1: Marginal effects of *placebo* on proportions of all coauthors in papers from CEPR and NBER

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Men and Women					Women			
	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 43,860$									
All	-0.0057 (0.0114)	-0.0046 (0.0098)	-0.0005 (0.0099)	0.0028 (0.0060)	0.0032 (0.0075)	-0.0009 (0.0048)	0.0018 (0.0052)	-0.0024 (0.0056)	-0.0007 (0.0036)
Men	-0.0077 (0.0116)	-0.0053 (0.0100)	-0.0011 (0.0101)	0.0057 (0.0063)	0.0028 (0.0064)	-0.0006 (0.0026)	0.0007 (0.0031)	-0.0019 (0.0041)	0.0005 (0.0033)
Women	0.0051 (0.0160)	-0.0010 (0.0137)	0.0021 (0.0137)	-0.0127* (0.0073)	0.0060 (0.0182)	-0.0024 (0.0203)	0.0067 (0.0171)	-0.0048 (0.0155)	-0.0065 (0.0060)
Junior authors, $N = 7,787$									
All	-0.0087 (0.0195)	0.0148 (0.0171)	0.0062 (0.0171)	-0.0057 (0.0095)	-0.0173 (0.0142)	-0.0117 (0.0085)	0.0033 (0.0092)	-0.0040 (0.0084)	-0.0036 (0.0048)
Men	-0.0031 (0.0202)	0.0070 (0.0170)	0.0048 (0.0178)	-0.0028 (0.0100)	-0.0152 (0.0111)	-0.0113 (0.0079)	0.0005 (0.0081)	-0.0035 (0.0043)	-0.0045 (0.0037)
Women	-0.0265 (0.0236)	0.0419* (0.0227)	0.0107 (0.0205)	-0.0159 (0.0108)	-0.0258 (0.0304)	-0.0127 (0.0128)	0.0116 (0.0142)	-0.0055 (0.0254)	-0.0003 (0.0093)
Senior authors, $N = 26,287$									
All	-0.0020 (0.0104)	-0.0083 (0.0094)	-0.0060 (0.0097)	0.0137* (0.0079)	0.0025 (0.0093)	0.0026 (0.0047)	-0.0055 (0.0050)	0.0002 (0.0062)	0.0021 (0.0046)
Men	-0.0042 (0.0105)	-0.0057 (0.0097)	-0.0079 (0.0099)	0.0158* (0.0081)	0.0027 (0.0083)	0.0028 (0.0027)	-0.0046 (0.0049)	-0.0012 (0.0059)	0.0034 (0.0046)
Women	0.0141 (0.0151)	-0.0261** (0.0113)	0.0078 (0.0142)	-0.0009 (0.0103)	0.0020 (0.0249)	0.0013 (0.0238)	-0.0107 (0.0080)	0.0083 (0.0104)	-0.0056 (0.0068)

Notes: The table presents the estimated marginal effects of a placebo MeToo indicator defined as the period between 1 July 2014 to 31 December 2016 on the nine outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table C.2: Marginal effects of *placebo* on proportions of new coauthors in papers from CEPR and NBER

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	New Men and Women					New Women				
	All	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 43,860$										
All	0.0203** (0.0089)	0.0061 (0.0073)	0.0005 (0.0052)	0.0036 (0.0074)	0.0028 (0.0056)	0.0054 (0.0065)	0.0005 (0.0030)	0.0006 (0.0031)	-0.0003 (0.0044)	-0.0009 (0.0033)
Men	0.0201** (0.0090)	0.0044 (0.0073)	0.0017 (0.0053)	0.0012 (0.0075)	0.0056 (0.0058)	0.0037 (0.0062)	-0.0003 (0.0027)	0.0003 (0.0029)	-0.0016 (0.0041)	-0.0001 (0.0032)
Women	0.0211* (0.0119)	0.0151 (0.0097)	-0.0056 (0.0068)	0.0157 (0.0096)	-0.0132** (0.0066)	0.0134 (0.0105)	0.0042 (0.0054)	0.0019 (0.0050)	0.0060 (0.0071)	-0.0048 (0.0049)
Junior authors, $N = 7,787$										
All	0.0256 (0.0211)	0.0189 (0.0184)	0.0164 (0.0145)	-0.0067 (0.0135)	-0.0061 (0.0091)	-0.0080 (0.0131)	-0.0045 (0.0070)	-0.0057 (0.0065)	0.0013 (0.0078)	-0.0051 (0.0050)
Men	0.0256 (0.0215)	0.0211 (0.0188)	0.0146 (0.0146)	-0.0084 (0.0140)	-0.0044 (0.0094)	-0.0114 (0.0117)	-0.0044 (0.0064)	-0.0059 (0.0059)	-0.0002 (0.0071)	-0.0061 (0.0038)
Women	0.0257 (0.0268)	0.0116 (0.0217)	0.0221 (0.0186)	-0.0010 (0.0163)	-0.0120 (0.0106)	0.0032 (0.0198)	-0.0044 (0.0103)	-0.0049 (0.0093)	0.0056 (0.0119)	-0.0014 (0.0095)
Senior authors, $N = 26,287$										
All	0.0267** (0.0120)	0.0080 (0.0083)	-0.0085 (0.0069)	0.0078 (0.0095)	0.0116 (0.0073)	0.0072 (0.0080)	0.0027 (0.0035)	-0.0026 (0.0037)	0.0019 (0.0059)	0.0020 (0.0042)
Men	0.0283** (0.0122)	0.0059 (0.0083)	-0.0075 (0.0071)	0.0080 (0.0097)	0.0144* (0.0076)	0.0071 (0.0078)	0.0015 (0.0031)	-0.0025 (0.0035)	0.0013 (0.0057)	0.0034 (0.0043)
Women	0.0155 (0.0165)	0.0223* (0.0120)	-0.0148* (0.0086)	0.0065 (0.0119)	-0.0076 (0.0088)	0.0079 (0.0136)	0.0098 (0.0072)	-0.0032 (0.0064)	0.0055 (0.0091)	-0.0058 (0.0060)

Notes: The table presents the estimated marginal effects of a placebo MeToo indicator defined as the period between 1 July 2014 to 31 December 2016 on the ten outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C.2 Time to Publication and the Timing of MeToo

We ran our analysis excluding identified longer-term papers. Our results remained largely consistent, with significance levels staying the same and effect sizes showing minimal changes. The corresponding estimates’ tables are in [Tables C.3](#) and [C.4](#). To contextualize these results, we benchmark them to the main findings we highlight in [Figure 4](#).

- Our first main result is that mixed-gender collaborations within the corresponding seniority group have increased.²⁹ When we exclude long-term papers, these effects remain largely consistent. The expected increase in the proportion of women coauthors for men remains at approximately the same level, with unchanged significance: it becomes slightly stronger when we look at all authors (by 1%) but weakens

²⁹We reach our first main result based on two pieces of evidence. First, our estimated increase in the proportion of women coauthors that men have in the full sample and when we focus on junior men with their junior women coauthors, and senior men with their corresponding senior women coauthors. Second, our estimated decrease in the proportion of women coauthors that women have when the MeToo indicator switches on, both when looking at all authors in the sample and within seniority groups.

when we zoom into senior and junior men (by 1% and 17% respectively). For women, we continue to find a decrease in collaborations with other women, but this effect strengthens when considering all authors (by 6%) and when we focus on senior women (by 3%). Significance (or lack of it for junior women) remains unchanged.

- Our second result highlights a decline in co-authorship between senior and junior economists.³⁰ When we exclude long-term papers from our sample, this result remains broadly unchanged in levels and significance. For junior authors, the decrease in the share of senior women coauthors weakens slightly (by 3%), while the change in their share of senior coauthors (including senior men) remains insignificant. For senior authors, we observe a further reduction in the share of junior coauthors as a whole and the share of junior women coauthors compared to the baseline specification (by 7–21%).
- Our last main finding suggests a general chilling effect on the expansion of economists' professional networks.³¹ When excluding long-term papers from our sample, this chilling effect weakens slightly when we consider all authors and when we focus on junior authors (by 7–9%). However, for senior authors, we observe a further decrease in new collaborations (by 9%) compared to our baseline specification.

Overall, excluding longer-term papers slightly strengthens the effects we find in our baseline specification, which is consistent with longer-term projects having a diluted immediate impact of MeToo on collaboration patterns.

We repeat our analysis excluding shorter-term papers and compare these estimates with our main findings from our baseline specification.

- In terms of our first finding — that mixed-gender collaborations have increased, — when we exclude faster papers, the effect diminishes when looking at all authors (by 18%), while it remains approximately the same for juniors and seniors (it strengthens by 3–9%). This could indicate that men were more likely to adjust their collaboration patterns quickly in response to MeToo in faster-turnaround papers. For women we see a further decrease in their share of male coauthors overall (by 26%) and for senior women (who see a 25% smaller share of senior men coauthors). As in the sample without slower papers, significance (or lack of it for junior women) remains unchanged. This potentially suggests a more cautious approach to collaboration in projects with longer commitments.
- Regarding our second finding — showing a decline in co-authorship between senior and junior economists, — we see that the expected changes when we exclude faster papers remain negative but are stronger for junior authors (by 10%) and slightly weaker for seniors (by 2–8%).
- Lastly, compared to our third finding — suggesting a general chilling effect on the expansion of economists' professional networks — when we exclude faster papers, we see that the decrease in new coauthorships across seniority groups becomes stronger overall and for junior authors (by 7 and 3% respectively) but weakens slightly for senior authors (by 5%).

We attempted to analyse faster papers and longer-term papers separately as well. However, both these subsamples proved too small to draw statistically meaningful conclusions.

³⁰In our baseline specification we estimate a decrease in the proportion of senior (and specifically senior women) coauthors that juniors have and, symmetrically, a decrease in the proportion of junior (and specifically junior women) coauthors that seniors have.

³¹We estimate decreases in the share of new coauthors (considering all seniorities and genders), the share of new senior coauthors for juniors, and the share of new junior coauthors for seniors in our baseline results.

Table C.3: Marginal effects of MeToo on proportions of all coauthors in papers that are not slow.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Men and Women						Women		
	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 43,044$									
All	-0.0181* (0.0094)	0.0176** (0.0084)	-0.0132* (0.0079)	0.0190*** (0.0054)	0.0029 (0.0063)	-0.0030 (0.0042)	0.0056 (0.0038)	-0.0048 (0.0047)	0.0068* (0.0035)
Men	-0.0206** (0.0096)	0.0219** (0.0088)	-0.0121 (0.0082)	0.0175*** (0.0055)	0.0128** (0.0056)	0.0047* (0.0026)	0.0079*** (0.0027)	-0.0011 (0.0036)	0.0066** (0.0033)
Women	-0.0070 (0.0150)	0.0003 (0.0124)	-0.0175 (0.0121)	0.0258*** (0.0083)	-0.0472*** (0.0161)	-0.0432** (0.0179)	-0.0022 (0.0126)	-0.0210 (0.0133)	0.0080 (0.0063)
Junior authors, $N = 7,597$									
All	-0.0232 (0.0175)	0.0004 (0.0128)	0.0137 (0.0145)	0.0121 (0.0087)	-0.0067 (0.0138)	-0.0254*** (0.0080)	0.0052 (0.0064)	0.0077 (0.0094)	-0.0010 (0.0044)
Men	-0.0241 (0.0181)	-0.0023 (0.0132)	0.0186 (0.0151)	0.0107 (0.0089)	0.0127 (0.0110)	-0.0192** (0.0078)	0.0056 (0.0062)	0.0145*** (0.0054)	-0.0007 (0.0040)
Women	-0.0199 (0.0233)	0.0091 (0.0172)	-0.0033 (0.0187)	0.0168 (0.0119)	-0.0798*** (0.0294)	-0.0449*** (0.0117)	0.0038 (0.0097)	-0.0159 (0.0261)	-0.0018 (0.0073)
Senior authors, $N = 25,879$									
All	-0.0122 (0.0086)	0.0038 (0.0075)	-0.0198** (0.0081)	0.0226*** (0.0068)	-0.0040 (0.0076)	-0.0015 (0.0042)	0.0026 (0.0044)	-0.0139*** (0.0044)	0.0097** (0.0044)
Men	-0.0109 (0.0088)	0.0023 (0.0077)	-0.0189** (0.0084)	0.0220*** (0.0070)	0.0062 (0.0071)	0.0066** (0.0027)	0.0026 (0.0043)	-0.0122*** (0.0044)	0.0098** (0.0043)
Women	-0.0193 (0.0138)	0.0119 (0.0114)	-0.0242** (0.0122)	0.0259** (0.0108)	-0.0712*** (0.0211)	-0.0499** (0.0206)	0.0024 (0.0083)	-0.0239*** (0.0072)	0.0095 (0.0073)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the nine outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table C.4: Marginal effects of MeToo on proportions of new coauthors in *papers that are not slow*.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	New Men and Women					New Women				
	All	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 43,044$										
All	-0.0170** (0.0070)	-0.0150*** (0.0053)	-0.0065 (0.0042)	-0.0096* (0.0055)	0.0185*** (0.0051)	-0.0039 (0.0049)	-0.0050** (0.0021)	0.0008 (0.0021)	-0.0053* (0.0032)	0.0056* (0.0031)
Men	-0.0184** (0.0073)	-0.0151*** (0.0054)	-0.0072* (0.0042)	-0.0085 (0.0057)	0.0171*** (0.0052)	-0.0008 (0.0048)	-0.0031 (0.0020)	0.0009 (0.0020)	-0.0038 (0.0031)	0.0055* (0.0030)
Women	-0.0106 (0.0103)	-0.0143* (0.0073)	-0.0036 (0.0058)	-0.0142* (0.0076)	0.0250*** (0.0076)	-0.0175** (0.0078)	-0.0132*** (0.0033)	0.0006 (0.0036)	-0.0120** (0.0050)	0.0058 (0.0049)
Junior authors, $N = 7,597$										
All	-0.0180 (0.0158)	-0.0271** (0.0134)	-0.0033 (0.0100)	0.0046 (0.0106)	0.0121 (0.0084)	-0.0141 (0.0094)	-0.0183*** (0.0066)	0.0011 (0.0047)	0.0004 (0.0059)	-0.0016 (0.0042)
Men	-0.0135 (0.0162)	-0.0275** (0.0139)	-0.0040 (0.0101)	0.0108 (0.0113)	0.0108 (0.0086)	-0.0052 (0.0092)	-0.0120** (0.0060)	0.0008 (0.0044)	0.0037 (0.0058)	-0.0010 (0.0039)
Women	-0.0333 (0.0221)	-0.0259 (0.0170)	-0.0009 (0.0141)	-0.0171 (0.0132)	0.0160 (0.0111)	-0.0417*** (0.0135)	-0.0418*** (0.0113)	0.0019 (0.0074)	-0.0099 (0.0082)	-0.0033 (0.0065)
Senior authors, $N = 25,879$										
All	-0.0189** (0.0096)	-0.0100 (0.0062)	-0.0074 (0.0056)	-0.0211*** (0.0068)	0.0218*** (0.0064)	-0.0080 (0.0057)	-0.0040* (0.0023)	0.0007 (0.0030)	-0.0128*** (0.0036)	0.0077** (0.0038)
Men	-0.0217** (0.0099)	-0.0102 (0.0063)	-0.0092 (0.0057)	-0.0211*** (0.0069)	0.0210*** (0.0066)	-0.0052 (0.0058)	-0.0024 (0.0023)	0.0006 (0.0029)	-0.0114*** (0.0036)	0.0078** (0.0037)
Women	-0.0028 (0.0147)	-0.0087 (0.0090)	0.0018 (0.0082)	-0.0202** (0.0099)	0.0261*** (0.0099)	-0.0233** (0.0094)	-0.0125*** (0.0042)	0.0012 (0.0054)	-0.0207*** (0.0058)	0.0075 (0.0064)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the ten outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table C.5: Marginal effects of MeToo on proportions of all coauthors in papers that are not fast.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Men and Women						Women		
	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 38,935$									
All	-0.0163* (0.0096)	0.0229*** (0.0088)	-0.0094 (0.0082)	0.0093* (0.0053)	-0.0006 (0.0067)	-0.0062 (0.0043)	0.0054 (0.0040)	-0.0019 (0.0050)	0.0016 (0.0032)
Men	-0.0184* (0.0099)	0.0270*** (0.0092)	-0.0081 (0.0085)	0.0079 (0.0054)	0.0104* (0.0059)	0.0040 (0.0027)	0.0077*** (0.0028)	0.0011 (0.0039)	0.0017 (0.0029)
Women	-0.0070 (0.0150)	0.0067 (0.0127)	-0.0148 (0.0122)	0.0155* (0.0082)	-0.0561*** (0.0165)	-0.0578*** (0.0172)	-0.0023 (0.0127)	-0.0142 (0.0137)	0.0012 (0.0059)
Junior authors, $N = 6,876$									
All	-0.0204 (0.0183)	-0.0003 (0.0134)	0.0193 (0.0151)	0.0033 (0.0086)	-0.0069 (0.0148)	-0.0286*** (0.0082)	0.0031 (0.0064)	0.0147 (0.0107)	-0.0032 (0.0044)
Men	-0.0186 (0.0191)	-0.0039 (0.0138)	0.0241 (0.0157)	0.0002 (0.0086)	0.0130 (0.0118)	-0.0215*** (0.0079)	0.0041 (0.0062)	0.0189*** (0.0063)	-0.0033 (0.0040)
Women	-0.0268 (0.0235)	0.0112 (0.0180)	0.0033 (0.0192)	0.0132 (0.0123)	-0.0835*** (0.0303)	-0.0507*** (0.0123)	-0.0004 (0.0099)	-0.0008 (0.0275)	-0.0029 (0.0074)
Senior authors, $N = 23,380$									
All	-0.0103 (0.0090)	0.0070 (0.0078)	-0.0159* (0.0085)	0.0121* (0.0066)	-0.0089 (0.0079)	-0.0029 (0.0044)	0.0021 (0.0045)	-0.0119*** (0.0046)	0.0029 (0.0039)
Men	-0.0088 (0.0092)	0.0049 (0.0080)	-0.0144 (0.0088)	0.0118* (0.0068)	0.0015 (0.0074)	0.0069** (0.0029)	0.0020 (0.0044)	-0.0107** (0.0046)	0.0031 (0.0037)
Women	-0.0185 (0.0142)	0.0181 (0.0119)	-0.0238* (0.0126)	0.0140 (0.0107)	-0.0780*** (0.0210)	-0.0604*** (0.0203)	0.0029 (0.0082)	-0.0190** (0.0077)	0.0024 (0.0071)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the nine outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

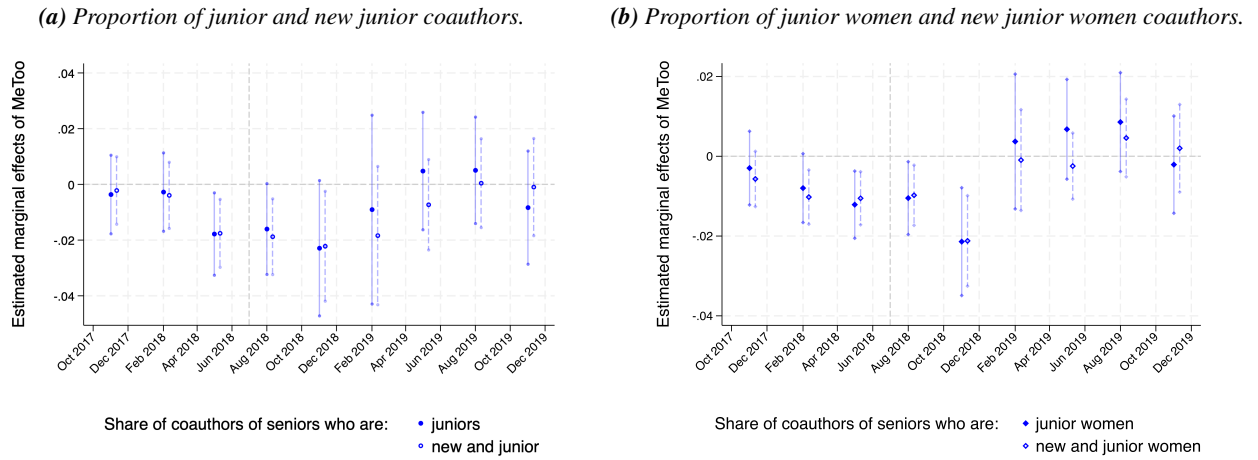
Table C.6: Marginal effects of MeToo on proportions of new coauthors in papers that are not fast.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	New Men and Women					New Women				
	All	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 38,935$										
All	-0.0194*** (0.0074)	-0.0152*** (0.0055)	-0.0050 (0.0044)	-0.0064 (0.0058)	0.0095* (0.0050)	-0.0057 (0.0051)	-0.0056** (0.0022)	0.0010 (0.0022)	-0.0031 (0.0035)	0.0008 (0.0029)
Men	-0.0210*** (0.0076)	-0.0152*** (0.0056)	-0.0057 (0.0045)	-0.0055 (0.0060)	0.0085* (0.0051)	-0.0030 (0.0050)	-0.0036* (0.0021)	0.0009 (0.0021)	-0.0019 (0.0034)	0.0009 (0.0028)
Women	-0.0123 (0.0108)	-0.0152** (0.0075)	-0.0020 (0.0062)	-0.0101 (0.0081)	0.0142* (0.0076)	-0.0169** (0.0082)	-0.0139*** (0.0036)	0.0015 (0.0039)	-0.0083 (0.0055)	0.0002 (0.0047)
Junior authors, $N = 6,876$										
All	-0.0256 (0.0166)	-0.0307** (0.0138)	-0.0021 (0.0107)	0.0075 (0.0112)	0.0026 (0.0082)	-0.0152 (0.0103)	-0.0212*** (0.0070)	-0.0005 (0.0048)	0.0063 (0.0069)	-0.0050 (0.0043)
Men	-0.0184 (0.0170)	-0.0281* (0.0146)	-0.0024 (0.0107)	0.0140 (0.0120)	0.0004 (0.0084)	-0.0047 (0.0100)	-0.0142** (0.0063)	-0.0003 (0.0045)	0.0105 (0.0069)	-0.0042 (0.0041)
Women	-0.0492** (0.0232)	-0.0396** (0.0168)	-0.0010 (0.0151)	-0.0149 (0.0137)	0.0095 (0.0112)	-0.0476*** (0.0143)	-0.0463*** (0.0122)	-0.0013 (0.0076)	-0.0067 (0.0090)	-0.0073 (0.0065)
Senior authors, $N = 23,380$										
All	-0.0233** (0.0101)	-0.0122* (0.0063)	-0.0048 (0.0059)	-0.0184*** (0.0071)	0.0122* (0.0063)	-0.0108* (0.0059)	-0.0045* (0.0024)	0.0026 (0.0033)	-0.0122*** (0.0038)	0.0021 (0.0034)
Men	-0.0262** (0.0104)	-0.0127* (0.0065)	-0.0070 (0.0060)	-0.0178** (0.0072)	0.0118* (0.0064)	-0.0091 (0.0060)	-0.0030 (0.0024)	0.0021 (0.0031)	-0.0112*** (0.0037)	0.0022 (0.0033)
Women	-0.0071 (0.0154)	-0.0094 (0.0091)	0.0068 (0.0090)	-0.0212** (0.0106)	0.0143 (0.0099)	-0.0200** (0.0099)	-0.0122*** (0.0043)	0.0049 (0.0060)	-0.0174*** (0.0062)	0.0016 (0.0063)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the ten outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C.3 Validating our choice of the MeToo data

Figure C.1: Predicted Marginal Effects for Different MeToo Start Dates for senior authors



Notes: The figure shows the estimated marginal effects of the MeToo indicator on the proportion of junior and new junior coauthors in Panel (a) and junior women and new junior women in Panel (b), for senior economists, with corresponding 95% confidence bands. The predicted shares are for senior economists as the estimates are based on the subsample containing only author-paper observations for senior economists. Each estimate is derived using a different starting point for the MeToo movement, ranging from October 2017 to December 2019, in three-month intervals. Solid dots represent the estimated marginal effects on the overall proportion of junior and junior women coauthors, while hollow dots indicate the estimated marginal effects on the proportion of new junior and new junior women coauthors.

C.4 Accounting for Gender-Specific Effects

The estimates from the specification with gender interactions broadly reflects our main findings of collaboration patterns post-MeToo as can be seen in [Tables C.7](#) and [C.8](#). We summarise the differences between this specification and our baseline specification by benchmarking the results we obtain with gender interactions to our main findings (summarised in [Figure 4](#) and described in *footnotes 29, 30, and 31* above for details on our main findings).

- Regarding our first result — that mixed-gender collaborations within the corresponding seniority group have increased, — the expected change in the proportion of women coauthors that men have when the MeToo indicator switches on, is reduced by 10–24% but remains significant across all samples. The decrease in women’s collaborations with other women is also reduced (by 15–24%) with unchanged significance.
- Our second result — highlighting a decline in co-authorship between senior and junior economists, — remains broadly unchanged in levels and significance. For juniors, the change in the share of senior coauthors overall remains insignificant, but significant for senior women coauthors. This effect is slightly weaker for men (by 10%) but stronger for women (by 4%). For the seniors, we see a reduction in the share of junior of coauthors as a whole and the share of junior women coauthors. This reductions, in contrast to the junior sample, are stronger for men (by 1%) and weaker for women (by 7–8%) with respect to the baseline specification.
- Our third main finding — suggesting a general chilling effect on the expansion of economists’ professional networks — remains surprisingly similar in levels and significance (variations ranging from 1 to 3%).

Table C.7: Marginal effects of MeToo on proportions of all coauthors when including sex interactions with all variables of author characteristics.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Men and Women						Women		
	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 43,860$									
Men	-0.0198** (0.0095)	0.0200** (0.0087)	-0.0082 (0.0082)	0.0146*** (0.0054)	0.0104* (0.0054)	0.0051* (0.0027)	0.0051** (0.0025)	-0.0024 (0.0034)	0.0052* (0.0031)
Women	-0.0160 (0.0148)	0.0020 (0.0121)	-0.0099 (0.0120)	0.0243*** (0.0086)	-0.0338** (0.0154)	-0.0451*** (0.0172)	0.0026 (0.0108)	-0.0072 (0.0111)	0.0066 (0.0062)
Junior authors, $N = 7,787$									
Men	-0.0201 (0.0177)	-0.0057 (0.0128)	0.0218 (0.0148)	0.0072 (0.0085)	0.0116 (0.0110)	-0.0183** (0.0075)	0.0051 (0.0060)	0.0156*** (0.0057)	-0.0017 (0.0038)
Women	-0.0158 (0.0232)	0.0045 (0.0170)	-0.0027 (0.0183)	0.0164 (0.0114)	-0.0705** (0.0292)	-0.0463*** (0.0111)	0.0033 (0.0095)	-0.0031 (0.0259)	-0.0014 (0.0074)
Senior authors, $N = 26,287$									
Men	-0.0116 (0.0086)	0.0019 (0.0075)	-0.0155* (0.0083)	0.0192*** (0.0068)	0.0039 (0.0069)	0.0051* (0.0026)	0.0020 (0.0042)	-0.0116*** (0.0043)	0.0087** (0.0041)
Women	-0.0242* (0.0141)	0.0155 (0.0116)	-0.0199* (0.0120)	0.0223** (0.0106)	-0.0543*** (0.0203)	-0.0409** (0.0196)	0.0074 (0.0087)	-0.0203*** (0.0069)	0.0081 (0.0076)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the nine outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table C.8: Marginal effects of MeToo on proportions of new coauthors when including sex interactions with all variables of author characteristics.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	New Men and Women					New Women				
	All	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 43,860$										
Men	-0.0198*** (0.0072)	-0.0166*** (0.0053)	-0.0077* (0.0042)	-0.0055 (0.0057)	0.0141*** (0.0051)	-0.0011 (0.0048)	-0.0033 (0.0020)	0.0005 (0.0020)	-0.0025 (0.0032)	0.0041 (0.0029)
Women	-0.0088 (0.0102)	-0.0153** (0.0070)	-0.0044 (0.0057)	-0.0106 (0.0076)	0.0233*** (0.0078)	-0.0152** (0.0077)	-0.0130*** (0.0032)	0.0015 (0.0038)	-0.0096** (0.0049)	0.0046 (0.0050)
Junior authors, $N = 7,787$										
Men	-0.0141 (0.0159)	-0.0288** (0.0136)	-0.0046 (0.0100)	0.0147 (0.0113)	0.0074 (0.0083)	-0.0033 (0.0095)	-0.0126** (0.0059)	0.0005 (0.0043)	0.0084 (0.0065)	-0.0024 (0.0038)
Women	-0.0398* (0.0216)	-0.0308* (0.0164)	-0.0043 (0.0141)	-0.0152 (0.0127)	0.0129 (0.0106)	-0.0464*** (0.0135)	-0.0456*** (0.0120)	0.0009 (0.0074)	-0.0091 (0.0080)	-0.0053 (0.0066)
Senior authors, $N = 26,287$										
Men	-0.0252** (0.0098)	-0.0132** (0.0062)	-0.0092 (0.0056)	-0.0196*** (0.0069)	0.0181*** (0.0065)	-0.0061 (0.0057)	-0.0029 (0.0023)	0.0007 (0.0029)	-0.0106*** (0.0036)	0.0063* (0.0036)
Women	-0.0007 (0.0145)	-0.0088 (0.0088)	0.0019 (0.0083)	-0.0173* (0.0098)	0.0235** (0.0100)	-0.0202** (0.0096)	-0.0116*** (0.0039)	0.0036 (0.0061)	-0.0184*** (0.0056)	0.0063 (0.0067)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the ten outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Appendix D. Heterogeneity Analysis

D.1 Differences between the NBER and CEPR

To re-estimate equation (2) while splitting the sample between NBER and CEPR working papers, we included duplicate papers from our initial dataset in both outlets. To illustrate the differences we find across the two outlets, we benchmark them to our main findings (summarised in [Figure 4](#)).

- Our first key finding is that mixed-gender collaborations within the corresponding seniority group have increased.³² This effect is particularly strong and statistically significant in the NBER dataset: the expected increase in the proportion of women coauthors for men increases further (27–33%). For women, we continue to find a decrease in collaborations with other women, but while this effect strengthens when considering all authors (by 29%) it loses significance when we zoom into the subsamples of senior and junior women. While the result remains present in CEPR papers, it loses statistical significance for men (overall and when we focus on junior and senior men), and for women and the subsample of junior women in the CEPR. However, CEPR senior women show a stronger reduction in collaborations with women (by 50%), as can be seen in [Figure 5, Panel \(a\)](#).
- Our second result highlights a decline in co-authorship between senior and junior economists.³³ Separately analysing the NBER and CEPR suggests that this effect is primarily driven by the NBER sample, where junior economists coauthor significantly less with senior authors (the decrease is 37–124% more pronounced than in our baseline results), though the decrease in collaborations NBER seniors have with juniors remains approximately the same (and loses significance when we look at their collaborations with senior women). In the CEPR sample, we find similar but weaker and statistically insignificant effects, for the most part. Notably, senior economists in the CEPR show a stronger reduction in collaborations with junior women (by 48%), as [Figure 5, Panel \(b\)](#) shows.
- Our last main finding suggests a general chilling effect on the expansion of economists’ professional networks.³⁴ While there is variation in the levels and significance of our findings across the two outlets, we continue to see a decrease in new collaborations. The decrease is stronger for junior collaborations with new seniors in the NBER dataset (by 27% with respect to the baseline). In the CEPR sample, the decrease is stronger when considering all authors collectively than in our baseline results (by 25%) and, while slightly weaker (by 10%), the decrease in new collaborations remains significant for senior with junior authors, as illustrated in [Figure 5, Panel \(c\)](#).

³²We detail how we reach this conclusion in [footnote 29](#).

³³See [footnote 30](#) for details.

³⁴Details of this finding are included in [footnote 31](#).

Table D.1: Marginal effects of MeToo on proportions of all coauthors in papers from NBER.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Men and Women						Women		
	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 30,718$									
All	-0.0073 (0.0108)	0.0095 (0.0095)	-0.0100 (0.0094)	0.0149** (0.0060)	0.0040 (0.0075)	-0.0046 (0.0051)	0.0003 (0.0042)	0.0019 (0.0058)	0.0069 (0.0053)
Men	-0.0117 (0.0111)	0.0146 (0.0100)	-0.0074 (0.0099)	0.0135** (0.0062)	0.0165** (0.0066)	0.0039 (0.0030)	0.0054* (0.0030)	0.0058 (0.0046)	0.0066* (0.0040)
Women	0.0116 (0.0172)	-0.0096 (0.0138)	-0.0201 (0.0136)	0.0211** (0.0092)	-0.0573*** (0.0189)	-0.0460** (0.0208)	-0.0188 (0.0134)	-0.0134 (0.0152)	0.0085 (0.0071)
Junior authors, $N = 5,894$									
All	-0.0424** (0.0189)	-0.0008 (0.0146)	0.0270* (0.0163)	0.0114 (0.0099)	-0.0060 (0.0161)	-0.0358*** (0.0095)	0.0039 (0.0073)	0.0117 (0.0115)	0.0028 (0.0055)
Men	-0.0453** (0.0197)	-0.0054 (0.0150)	0.0377** (0.0170)	0.0083 (0.0099)	0.0159 (0.0134)	-0.0281*** (0.0091)	0.0038 (0.0069)	0.0232*** (0.0073)	0.0021 (0.0049)
Women	-0.0333 (0.0251)	0.0128 (0.0202)	-0.0065 (0.0210)	0.0206 (0.0143)	-0.0825** (0.0321)	-0.0587*** (0.0141)	0.0040 (0.0120)	-0.0223 (0.0286)	0.0046 (0.0097)
Senior authors, $N = 18,165$									
All	-0.0056 (0.0099)	0.0058 (0.0089)	-0.0167* (0.0099)	0.0127* (0.0074)	0.0002 (0.0090)	0.0014 (0.0053)	-0.0023 (0.0046)	-0.0060 (0.0054)	0.0063 (0.0047)
Men	-0.0059 (0.0102)	0.0036 (0.0090)	-0.0138 (0.0104)	0.0127* (0.0075)	0.0109 (0.0083)	0.0085** (0.0034)	-0.0025 (0.0045)	-0.0038 (0.0055)	0.0074 (0.0052)
Women	-0.0038 (0.0161)	0.0177 (0.0141)	-0.0332** (0.0136)	0.0131 (0.0114)	-0.0694*** (0.0259)	-0.0368 (0.0247)	-0.0015 (0.0093)	-0.0183** (0.0078)	0.0008 (0.0072)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the nine outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.2: Marginal effects of MeToo on proportions of new coauthors in papers from NBER.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	New Men and Women					New Women				
	All	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 30,718$										
All	-0.0110 (0.0085)	-0.0131** (0.0059)	-0.0057 (0.0047)	-0.0035 (0.0067)	0.0142** (0.0058)	-0.0027 (0.0059)	-0.0051** (0.0020)	-0.0017 (0.0023)	-0.0009 (0.0041)	0.0053 (0.0038)
Men	-0.0115 (0.0088)	-0.0139** (0.0061)	-0.0065 (0.0047)	-0.0007 (0.0070)	0.0132** (0.0060)	0.0009 (0.0058)	-0.0034 (0.0023)	-0.0018 (0.0021)	0.0012 (0.0041)	0.0057 (0.0037)
Women	-0.0089 (0.0122)	-0.0098 (0.0083)	-0.0029 (0.0070)	-0.0151* (0.0089)	0.0188** (0.0083)	-0.0175* (0.0092)	-0.0120*** (0.0037)	-0.0017 (0.0044)	-0.0096 (0.0061)	0.0042 (0.0055)
Junior authors, $N = 5,894$										
All	-0.0051 (0.0178)	-0.0379** (0.0149)	0.0011 (0.0116)	0.0166 (0.0124)	0.0135 (0.0096)	-0.0098 (0.0116)	-0.0226*** (0.0080)	-0.0027 (0.0051)	0.0073 (0.0079)	0.0020 (0.0053)
Men	0.0048 (0.0182)	-0.0364** (0.0155)	0.0005 (0.0115)	0.0272** (0.0134)	0.0113 (0.0099)	0.0011 (0.0114)	-0.0144** (0.0070)	-0.0034 (0.0044)	0.0129 (0.0083)	0.0022 (0.0049)
Women	-0.0354 (0.0250)	-0.0426** (0.0186)	0.0031 (0.0174)	-0.0163 (0.0148)	0.0202 (0.0130)	-0.0414*** (0.0160)	-0.0513*** (0.0143)	-0.0009 (0.0093)	-0.0083 (0.0095)	0.0015 (0.0083)
Senior authors, $N = 18,165$										
All	-0.0118 (0.0113)	-0.0061 (0.0071)	-0.0051 (0.0065)	-0.0145* (0.0081)	0.0133* (0.0071)	-0.0069 (0.0067)	-0.0022 (0.0025)	-0.0022 (0.0031)	-0.0077* (0.0044)	0.0047 (0.0042)
Men	-0.0151 (0.0116)	-0.0068 (0.0073)	-0.0077 (0.0064)	-0.0135 (0.0084)	0.0127* (0.0073)	-0.0044 (0.0068)	-0.0011 (0.0026)	-0.0028 (0.0030)	-0.0061 (0.0045)	0.0054 (0.0042)
Women	0.0060 (0.0176)	-0.0026 (0.0105)	0.0081 (0.0102)	-0.0205* (0.0109)	0.0167 (0.0107)	-0.0199* (0.0110)	-0.0079** (0.0039)	0.0002 (0.0059)	-0.0164** (0.0064)	0.0017 (0.0068)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the ten outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.3: Marginal effects of MeToo on proportions of all coauthors in papers from *CEPR*.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
		Men and Women					Women		
	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 18,631$									
all	-0.0464*** (0.0143)	0.0342*** (0.0130)	0.0042 (0.0119)	0.0172** (0.0084)	-0.0027 (0.0094)	-0.0024 (0.0059)	0.0122** (0.0059)	-0.0086 (0.0073)	0.0006 (0.0045)
Men	-0.0463*** (0.0146)	0.0363*** (0.0135)	0.0036 (0.0121)	0.0159* (0.0085)	0.0021 (0.0083)	0.0046 (0.0039)	0.0093** (0.0042)	-0.0064 (0.0053)	0.0005 (0.0041)
Women	-0.0473** (0.0223)	0.0243 (0.0194)	0.0070 (0.0194)	0.0242* (0.0128)	-0.0301 (0.0243)	-0.0438* (0.0253)	0.0292 (0.0207)	-0.0205 (0.0221)	0.0010 (0.0095)
Junior authors, $N = 2,851$									
All	0.0023 (0.0281)	0.0026 (0.0201)	0.0204 (0.0227)	0.0016 (0.0117)	-0.0183 (0.0211)	-0.0130 (0.0125)	0.0019 (0.0088)	0.0090 (0.0147)	-0.0146 (0.0097)
Men	0.0057 (0.0286)	0.0022 (0.0207)	0.0174 (0.0234)	0.0000 (0.0120)	0.0001 (0.0159)	-0.0089 (0.0121)	0.0024 (0.0085)	0.0086 (0.0068)	-0.0147* (0.0088)
Women	-0.0156 (0.0402)	0.0043 (0.0276)	0.0360 (0.0299)	0.0095 (0.0161)	-0.1061** (0.0489)	-0.0305 (0.0189)	-0.0010 (0.0137)	0.0025 (0.0459)	-0.0124 (0.0132)
Senior authors, $N = 11,550$									
All	-0.0270** (0.0130)	0.0020 (0.0113)	-0.0084 (0.0119)	0.0281*** (0.0109)	-0.0158 (0.0114)	-0.0067 (0.0054)	0.0081 (0.0072)	-0.0192*** (0.0066)	0.0067 (0.0061)
Men	-0.0236* (0.0134)	-0.0011 (0.0116)	-0.0094 (0.0121)	0.0273** (0.0113)	-0.0083 (0.0107)	0.0033 (0.0037)	0.0071 (0.0070)	-0.0188*** (0.0065)	0.0050 (0.0060)
Women	-0.0462** (0.0198)	0.0192 (0.0167)	-0.0030 (0.0189)	0.0328* (0.0172)	-0.0673** (0.0295)	-0.0723*** (0.0272)	0.0143 (0.0130)	-0.0235* (0.0125)	0.0157 (0.0117)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the nine outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

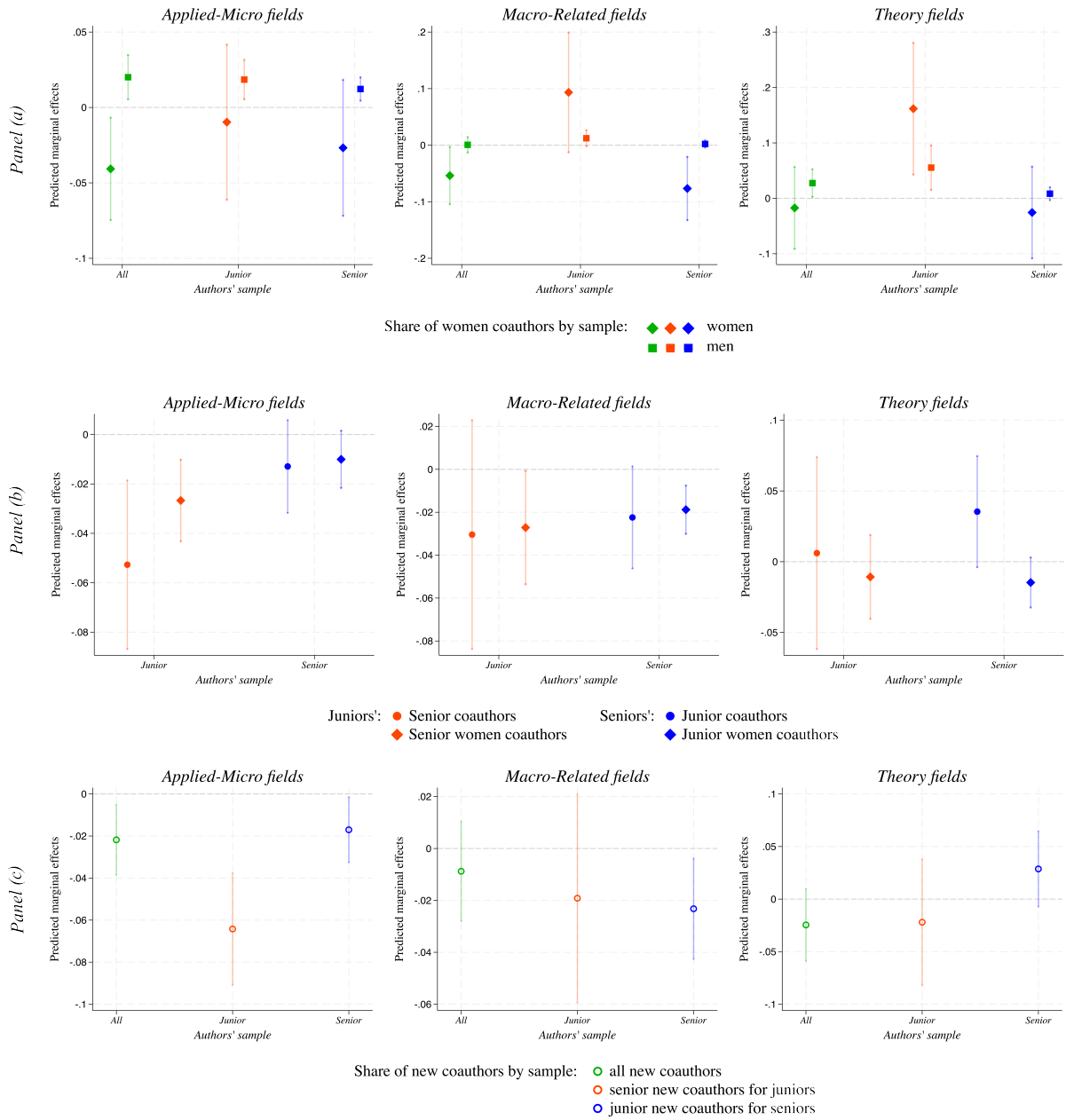
Table D.4: Marginal effects of MeToo on proportions of new coauthors in papers from *CEPR*.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	New Men and Women					New Women				
	All	Senior	Midcareer	Junior	Early career	All	Senior	Midcareer	Junior	Early career
All authors, $N = 18,631$										
All	-0.0227** (0.0106)	-0.0224*** (0.0085)	-0.0071 (0.0065)	-0.0035 (0.0083)	0.0179** (0.0079)	-0.0089 (0.0071)	-0.0063* (0.0036)	0.0047 (0.0037)	-0.0065 (0.0046)	0.0002 (0.0037)
Men	-0.0254** (0.0109)	-0.0217** (0.0086)	-0.0079 (0.0068)	-0.0047 (0.0084)	0.0162** (0.0079)	-0.0085 (0.0069)	-0.0048 (0.0034)	0.0041 (0.0036)	-0.0061 (0.0043)	-0.0004 (0.0035)
Women	-0.0090 (0.0158)	-0.0254** (0.0115)	-0.0032 (0.0082)	0.0027 (0.0125)	0.0271** (0.0120)	-0.0113 (0.0125)	-0.0139** (0.0062)	0.0078 (0.0059)	-0.0085 (0.0084)	0.0034 (0.0070)
Junior authors, $N = 2,851$										
All	-0.0310 (0.0263)	-0.0079 (0.0228)	-0.0061 (0.0167)	0.0084 (0.0173)	-0.0031 (0.0114)	-0.0300** (0.0145)	-0.0165 (0.0115)	0.0087 (0.0083)	-0.0069 (0.0084)	-0.0216 (0.0143)
Men	-0.0348 (0.0269)	-0.0114 (0.0234)	-0.0071 (0.0174)	0.0092 (0.0176)	-0.0041 (0.0116)	-0.0232* (0.0139)	-0.0133 (0.0110)	0.0081 (0.0082)	-0.0046 (0.0076)	-0.0210 (0.0135)
Women	-0.0126 (0.0372)	0.0089 (0.0304)	-0.0006 (0.0207)	0.0035 (0.0234)	0.0020 (0.0156)	-0.0582*** (0.0217)	-0.0357* (0.0186)	0.0104 (0.0124)	-0.0154 (0.0138)	-0.0210 (0.0173)
Senior authors, $N = 11,550$										
All	-0.0319** (0.0148)	-0.0232** (0.0095)	-0.0122 (0.0086)	-0.0173* (0.0104)	0.0257** (0.0101)	-0.0144* (0.0087)	-0.0097** (0.0042)	0.0065 (0.0057)	-0.0152*** (0.0055)	0.0047 (0.0048)
Men	-0.0337** (0.0150)	-0.0233** (0.0096)	-0.0130 (0.0089)	-0.0185* (0.0104)	0.0246** (0.0104)	-0.0134 (0.0086)	-0.0083** (0.0039)	0.0064 (0.0055)	-0.0144*** (0.0052)	0.0037 (0.0046)
Women	-0.0211 (0.0223)	-0.0216 (0.0136)	-0.0075 (0.0115)	-0.0106 (0.0164)	0.0318** (0.0156)	-0.0212 (0.0157)	-0.0179** (0.0081)	0.0078 (0.0102)	-0.0207** (0.0102)	0.0104 (0.0099)

Notes: The table presents the estimated marginal effects of the MeToo indicator on the ten outcomes listed in the table header. Each panel corresponds to one of our samples: the full sample (“All Authors”), junior authors (“Junior”), and senior authors (“Senior”). Within each panel, the table includes three rows of estimates: (i) the first row shows the estimated marginal effects from our preferred specification without gender interactions. (ii) The second and third rows report the estimated marginal effects from our preferred specification with gender interactions, calculated separately for male and female authors respectively. Standard errors, clustered by author, are reported in parentheses. Statistical significance is indicated by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

D.2 Heterogeneity across Research Fields

Figure D.1: Predicted Marginal Effects on Cross-gender Collaborations by Field



Notes: The panels in the figure show the estimated marginal effects of the MeToo period indicator from equation (2) with corresponding 90% confidence bands for three different subsamples: (i) our baseline, (ii) our sample excluding long-term papers, and (iii) our sample excluding faster papers. These effects capture the expected change in the proportion of specific types of coauthors when the MeToo indicator switches on, assuming all other factors remain constant. *Panel (a)* at the top displays estimates for the proportion of women coauthors. Green estimates correspond to the whole sample, orange to junior economists, and blue to senior economists. *Panel (b)* in the middle focuses on junior and senior economists. Orange estimates represent junior economists, while blue represents senior economists. *Panel (c)* at the bottom shows estimates for new coauthorships. Green estimates represent the share of new coauthors in the whole sample, orange represents new senior coauthors for junior economists, and blue represents new junior coauthors for senior economists.